Workload Reduction and Retention of Experienced Workers

Elena Castellaro*

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Abstract

Population aging is pushing governments to extend working lives, yet workers in demanding occupations often leave their jobs well before any retirement age binds. This paper examines whether reducing workload can keep them in their jobs longer. I study a Dutch program that subsidizes reduced teaching loads for older secondary-school teachers, exploiting a 2014 reform that restricted eligibility by date of birth in a regression discontinuity design. Using administrative records on the universe of Dutch secondary-school teachers, I find that eligibility raises take-up but has no significant effect on retention. This null conceals opposing responses by contract size. Full-time teachers are 6.1 percentage points more likely to continue teaching, and the teaching they provide by staying offsets the reduction in their teaching load. Part-time teachers, who already work reduced schedules, take up the subsidized reduction without staying longer, resulting in a net loss of teaching hours. Eligibility also appears to improve physical health, with linked health-insurance claims showing a reduction in pain-medication prescriptions. Workload reduction can thus ease strain but retains only workers for whom reducing hours is otherwise costly.

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1 Introduction

Population aging is pushing governments to raise retirement ages, but longer working lives may be difficult to sustain in physically or mentally demanding jobs. Prolonged work-related strain accelerates health decline (Ravesteijn et al., 2018) and drives workers out of their occupation well before any retirement age binds (Filer and Petri, 1988; Hayward et al., 1989; Blundell et al., 2023). Keeping these workers may therefore require changing the conditions of work, not only the age at which it ends. This paper examines whether reducing workload at later career stages retains workers in a demanding profession.

I study this question in Dutch secondary education, exploiting a national program (BAPO, "Promotion of Labor Participation of Older Workers") that allowed teachers aged 52 and older to reduce their annual teaching load by up to 10 percent. The program covered roughly 65 percent of the forgone salary, while teachers retained full pension rights. Teaching is a natural setting for this question: workload is a major source of teacher stress, and teacher shortages are widespread across OECD countries (OECD, 2025, 2023). The Netherlands faces similar challenges. Among workers aged 50–65, teachers report burnout complaints nearly twice as often as other workers (30 versus 17 percent; Netherlands Working Conditions Survey, NEA), and workload is one of the main reasons teachers give for leaving (Van Casteren et al., 2023). Retaining teachers is particularly important in a sector where outflows persistently exceed inflows (Adriaens and den Uijl, 2025) and unfilled vacancies are often covered by substitute teachers without the relevant qualifications, potentially affecting instructional quality (Herrmann and Rockoff, 2012; Benhenda, 2022).

A subsidized reduction in teaching hours may retain teachers who would otherwise leave the profession because of high workload. Although Dutch teachers can reduce their contracted hours without a subsidy, doing so is costly: they forgo wages and pension accrual, may be unable to regain those hours later, and may have their request denied. The subsidy should therefore matter most for teachers who would prefer fewer hours but have not reduced them because of these costs, and less for teachers who already work the hours they prefer. Contract size provides a proxy for this distinction: part-time teachers are more likely to have adjusted their hours to their preferences, whereas full-time teachers may include workers who would prefer fewer hours but have not made that adjustment. Consistent with this interpretation, late-career full-time teachers report greater work pressure than their part-time counterparts and are more likely to say they would work to a higher age if they could work fewer hours (Netherlands Working Conditions Survey, NEA: Table C.1).

Estimating the causal effect of workload reduction on retention is complicated by voluntary program participation. Participants may differ from non-participants in job strain, health, and attachment to the profession, which can independently affect retention. To address this selection problem, I exploit a 2014 reform that raised the eligibility age for subsidized workload reductions from 52 to 57, with the new age applying only to teachers born after a cutoff date. Teachers born just before the cutoff (the treatment group) remained eligible from age 52, whereas those born just after it (the control group) became eligible only at 57. I use a regression discontinuity (RD) design around the birth-date cutoff to estimate the causal effect of access to the program from age 52 rather than 57, over the years before the control group becomes eligible. Throughout, I refer to this early access as eligibility.

The analysis draws on administrative data covering the universe of teachers in Dutch secondary schools. Individual identifiers link employment records to demographic characteristics, labor-market outcomes, and mandatory health-insurance claims. The claims include antidepressant and pain-medication use, mental healthcare utilization, and total healthcare expenditure. These outcomes capture mental health, physical health, and overall healthcare use. Pain medication is of particular interest because teachers commonly report that their job harms their physical health (OECD, 2020), and musculoskeletal problems are among the most prevalent occupational health complaints in the profession (Erick and Smith, 2011).

Three findings emerge. First, eligibility increases BAPO take-up by 40 percentage points but has no detectable effect on retention in the full sample, with a pooled estimate of 1.0 percentage point that is not statistically significant. This aggregate result conceals substantial heterogeneity by contract type. Among full-time teachers, eligibility increases the probability of remaining in teaching over 2015–2018 by 6.1 percentage points, a 6.8 percent increase relative to the control mean at the cutoff. Among part-time teachers, the estimated effect is negative and statistically insignificant. The full-time effect appears in the first year of eligibility and persists through 2024, after the control cohort has itself become eligible at 57, so the teachers lost during the five-year deferral are not recovered. Estimates by destination are imprecise but suggest that most of the retained teachers would otherwise have moved to another job, largely outside education, with the remainder no longer in salaried employment.

Second, the retention response determines the program's effect on total teaching time. In the full sample, the estimated effect on total teaching supplied over 2016-2018 is negative but statistically insignificant, with the confidence interval ruling out any meaningful

increase. This average combines two opposing responses. Among full-time teachers, the point estimate is positive, and the confidence interval rules out a decline greater than 0.023 FTE, indicating that the additional teaching supplied by retained teachers offsets the hours released by the program. Among part-time teachers, by contrast, no retention response offsets the leave, and total teaching supplied falls by 0.067 FTE. Both patterns persist through 2024.

Third, reducing teaching hours appears to improve physical health. Eligibility reduces pain-medication use by 5.7 percentage points, a 27 percent decrease relative to the control mean, with no detectable effects on antidepressant use or healthcare expenditure. The effect is clearest among part-time teachers, whose retention does not change, so the estimate largely isolates the effect of fewer hours. Among full-time teachers, who both work fewer hours and stay longer, no health outcome changes significantly; the longer careers come at no detectable health cost on these measures, though the estimates are too imprecise to rule out moderate ones.

Two mechanisms could explain why eligibility retains full-time teachers. Fewer hours may make a demanding job more sustainable, or teachers may value the subsidized time off itself. The two differ in one prediction: if leave is valued in itself, workers should stay regardless of strain. I apply the same design to full-time support staff, who are entitled to the same program but report far less work pressure, and compare them with full-time teachers in the lowest salary scale, whose pay is closer to the support-staff. Take-up rises by more than 30 percentage points in both groups, but retention rises only among these teachers, and more strongly than in the full-time sample as a whole, consistent with the program retaining workers by relieving strain rather than by offering cheap leisure.

When an eligible teacher stays but takes leave, her school must cover a fraction of her timetable rather than a full vacancy. Exploiting cross-school variation in exposure to eligible cohorts, I find that the hours seniors give up are covered by younger incumbents expanding their contracts, with no detectable change in hiring.

The main benefit of retaining a teacher is avoiding a vacancy, which in Dutch secondary schools is mostly covered by teachers not qualified in the subject. Using estimates from the teacher value-added literature to value the resulting loss in student achievement, I find that the program yields a positive net return overall. The gain comes entirely from full-time teachers, for whom an additional teacher-year retained costs about half an annual salary. Among part-time teachers, the subsidy reduces teaching supplied without raising retention, and the net return is negative.

Related literature This paper contributes to three literatures. The first concerns policies that lower the cost of reducing hours late in the career. Because hours might be costly to adjust within a job (Altonji and Paxson, 1988, 1992), older workers who want shorter hours often leave rather than reduce them (Ameriks et al., 2020; Gielen, 2009), and subsidized reductions in working time aim to keep them employed by lowering this cost. Evaluations close to the retirement margin find that participants cut their hours while retiring no later, or only temporarily later, so that total labor supply is unchanged or falls (Graf et al., 2011; Albanese et al., 2020; Börsch-Supan et al., 2018); a mandated extra week of leave at 60 has no effect on employment at all (Bensnes et al., 2026); and a positive net effect has been found in one setting, Dutch municipal gradual-retirement plans (Rutten et al., 2025). Börsch-Supan et al. (2018) show that the net effect depends on how many participants are already at their preferred hours, which they cannot observe. I study an earlier margin, at which the alternative to a demanding job is usually another job rather than retirement, in an occupation where hours are the main lever on workload. Dutch teachers can reduce their hours, but at a cost in pay, pension accrual, and the school's consent. Using the pre-reform contract to distinguish teachers who have already paid this cost from those who have not, I find the same aggregate null, but it conceals a retention gain among full-time teachers.

The second evaluates policies that affect teacher turnover. Most causal evidence on retention policies concerns compensation: higher pay lowers turnover (Clotfelter et al., 2008; Falch, 2011; Hendricks, 2014) and shapes sorting across districts (Biasi, 2021); pension incentives govern late-career exit (Brown, 2013); and targeted bonuses raise early-career retention (Sims and Benhenda, 2022). Few studies evaluate changes to working conditions causally. The two studies of late-career workload reductions for teachers, a free 5.8 percent cut in teaching load at age 55 in Norway (Bratberg et al., 2020) and the pre-reform BAPO in the Netherlands (Leuven, 2006), examine sickness absence rather than retention. I instead estimate the effect on retention and identify it from a reform rather than an age threshold, comparing teachers of the same age who differ only in access to the reduction.

The third asks whether shorter hours improve health. Evidence that reducing hours improves health comes mainly from statutory workweek reductions covering the whole workforce (Ahn, 2016; Sánchez, 2017; Cygan-Rehm and Wunder, 2018; Berniell and Bietenbeck, 2020), which find at most modest effects on self-reported health. Because these reforms cut contractual hours without changing the work itself, they cannot separate fewer hours from a lighter workload. Like my setting, Bratberg et al. (2020) isolate the

effect of workload through a direct cut in lessons for Norwegian teachers at 55. They find lower sickness absence and better mental health, concentrated among men in poor prior health, and no effect on musculoskeletal problems. I instead find an improvement in musculoskeletal health, proxied by pain-medication use. Because retention does not respond among part-time teachers, the estimate for that group isolates the effect of a lighter workload.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting and the 2014 reform. Section 3 presents the data. Section 4 discusses the empirical strategy and its validation. Section 5 reports the main labor-market results, and Section 6 the health results. Section 7 sheds light on the mechanisms. Section 8 presents the robustness checks. Section 9 traces spillovers within schools and Section 10 presents a discussion on the cost-benefit analysis of the subsidized leave. Section 11 concludes.

2 Institutional Context

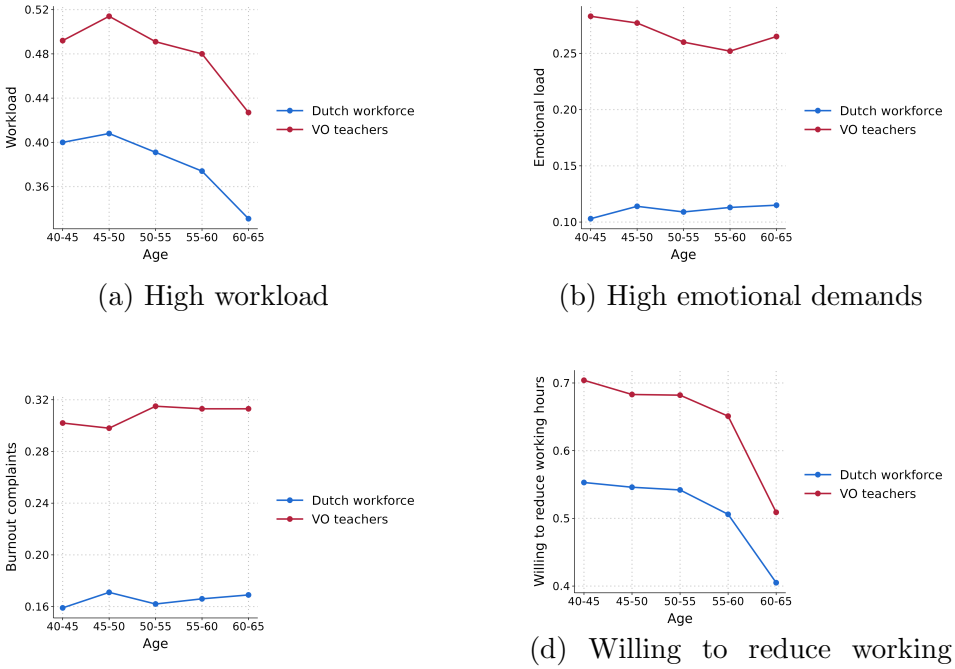
2.1 Dutch Secondary Education

Dutch secondary education (*voortgezet onderwijs*, VO) consists of three tracks that differ in length and in the tertiary education they give access to: VMBO (four years, pre-vocational), HAVO (five years, leading to universities of applied sciences), and VWO (six years, leading to research universities). Schools recruit and employ staff autonomously, but wages, pay scales, and working-time arrangements are set sector-wide through the collective labor agreement (CAO-VO), negotiated between school boards and unions. Personnel is formally divided into management, teaching staff, and support staff, a legal distinction grounded in qualification requirements. Teaching staff hold a statutory teaching qualification and bear formal responsibility for the classroom, whereas support staff face no qualification requirement and sit on separate salary scales.[1] In addition to directly employed teaching staff, schools hire teachers through temporary agencies or as self-employed contractors. Schools use this channel mainly to cover sickness absence and unplanned vacancies and it accounted for 2.3 percent of personnel costs in 2013 and 3.7 percent in 2015 (Leemans et al., 2017). For teaching staff, a full-time appointment corresponds to a 40-hour week, of which 22 hours are spent teaching in 50-minute periods; the remainder covers preparation, grading, and meetings. Teachers retain considerable control over their working time, as they may request annual adjustments to their contractual

[1]Support staff include teaching assistants, technical and ICT staff, and administration. Teaching assistants may work directly with classes, but only under the responsibility of a qualified teacher.

full-time equivalent (FTE). Part-time work is therefore widespread: the average teacher held a 0.83 FTE contract in 2013 (Table 1). Despite this flexibility, the sector appears to be under considerable strain. Data from the Netherlands Working Conditions Survey (NEA; see Appendix C) show that, compared with the Dutch workforce as a whole, VO teachers are more likely to report high workload, high emotional demands, and burnout complaints, and more likely to want to reduce their working hours in order to keep working for more years (Figure 1). These differences are present in every age group between 50 and 65.

Figure 1: Work-related strain by age group, secondary-school teachers versus the Dutch workforce



(a) High workload

(b) High emotional demands

(c) Burnout complaints

(d) Willing to reduce working hours

Notes: Each panel shows the share of employees reporting the outcome, by age group, for secondary-school (VO) teachers and for the Dutch workforce as a whole. Variable definitions are given in appendix C. Source: Netherlands Working Conditions Survey (NEA), 2014–2024.

2.2 BAPO and the 2014 Reform

Since 1994, teachers aged 52 and older have been eligible for BAPO (*Bevordering Arbeidsparticipatie Ouderen*, "Promotion of labor Participation of Older Workers"), a subsi-

dized leave program that allowed them to reduce their working hours at a reduced cost. Schools subsidized 65% of the salary associated with the leave, while pension accrual continued on the full contract. Teachers could also reduce their hours without BAPO, but this lowered both salary and pension accrual and could make hours difficult to regain.[2]

Schools financed the subsidy from their lump-sum grant, which varies with enrolment but not with the age of staff.[3] As the workforce aged, program costs increased and the scheme came under political pressure.[4] The 2014 reform of the CAO-VO responded by raising the minimum age for subsidized leave from 52 to 57. The reform was announced in April 2014 and took effect on 1 August of that year. It included a transitional arrangement, which provides the treatment studied here: teachers born before 1 August 1962, who had turned 52 by the 31 July 2014 reference date and had been continuously employed by a publicly funded educational or research institution since 1 August 2009, retained access to subsidized leave from age 52. Teachers born on or after 1 August 1962 became eligible only at age 57, a five-year deferral.[5] Figure 2 plots the eligibility profiles of the two cohorts (blue and orange lines). I exploit this discontinuity in a regression discontinuity design, comparing teachers born just before (treated) and just after the cutoff (control) to estimate the causal effect of access to the program during the five-year window.

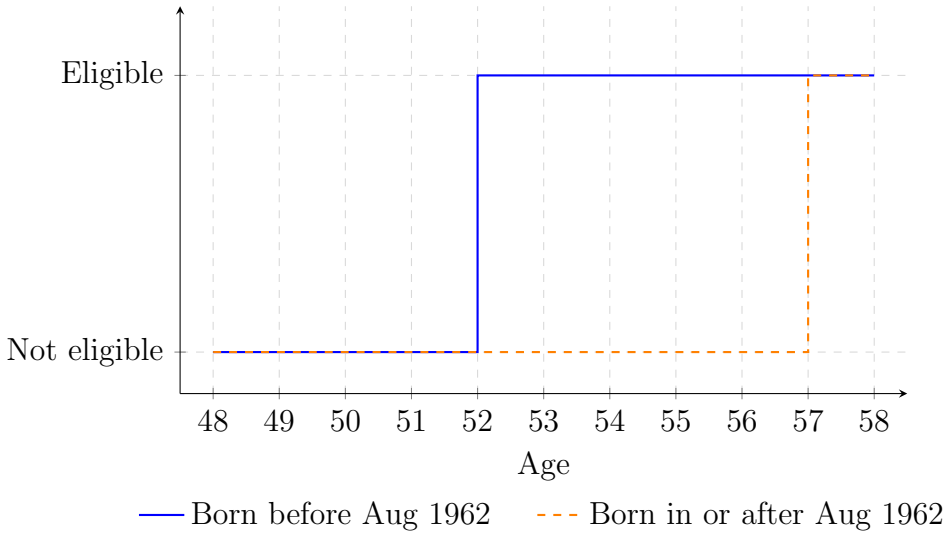
[2]Teachers may request a contract adjustment once a year (VO-raad, 2014, art. 6.1), but increases may be refused due to a lack of work, formation space, or personnel budget (*Wet aanpassing arbeidsduur*, art. 2).

[3]The grant's personnel component multiplies a pupil-driven number of normative staff positions by a flat national wage norm per school type and contains no BAPO component (Algemene Rekenkamer, 2014).

[4]A 2011 parliamentary motion urged that the scheme be phased out (Tweede Kamer, 2011), and an assessment of the teacher labor market found high take-up but little reduction in sickness absence (Leuven, 2006).

[5]From age 57, teachers were allowed to obtain an additional 170 hours of leave under certain circumstances.

Figure 2: Eligibility for the BAPO leave program after 1 August 2014



Under the transitional arrangement, a full-time teacher in the treatment group was entitled to 50 hours of leave without co-payment and 120 hours with a 50 percent co-payment, for 170 hours in total.[6] This amounts to about 10 percent of the 1,659 annual working hours of a full-time contract and corresponds to a reduction of three 50-minute lessons per week (VO-raad, 2014).[7] The leave entitlement scales with contract size, as do all rights under the CAO-VO (VO-raad, 2014), while pension accrual and other salary-linked entitlements are based on the unreduced salary. BAPO leave cannot fall below the previous year's amount at a constant contract size and to ensure that the leave is not taxed as early retirement, a teacher must continue working at least half of their pre-leave weekly hours (VO-raad, 2014).[8]

Teachers in the control group had no access to subsidized leave before 57. All they received after the 2014 reform was the 50-hour *basisbudget* granted to every employee,

[6]For support staff, the co-payment was 40%

[7]For a full-time teacher taking all 170 hours, the co-payment is $0.5 \times 120/1{,}659 \approx 3.6$ percent of annual salary. The price of leave was essentially unchanged: under the old scheme, with a 35 percent co-payment on all 170 hours, leave hours were paid at 65 percent of salary; under the transitional arrangement, the corresponding rate was $(50 + 0.5 \times 120)/170 \approx 0.65$.

[8]BAPO alone never breaches this floor, which binds only in combination with other leave. Since 2005, an employer has owed a final levy (52 percent from 2011) on payments under schemes that bridge income to the pension date; reduced-hours arrangements for older workers escape it only if the employee continues working at least half of the original contract (*Wet op de loonbelasting 1964*, art. 32ba; *Uitvoeringsregeling loonbelasting 2011*, art. 8.7).

which could be used for task reallocation, training, or leave (VO-raad, 2014).[9] The difference between the cohorts at age 52 is thus the 120 co-paid hours, scaled to contract size. The transitional arrangement and *basisbudget* differ in usability, however. BAPO hours are taken as leave, whereas use of the *basisbudget* depends on school-level task allocation: among staff who do not use it for training or task reallocation, 52 percent report that their school offers no opportunity to take the hours, while school leaders cite lesson coverage and limited replacement capacity as barriers (Voion, 2026). I therefore interpret the estimates as the effect of the BAPO package as a whole rather than of 120 hours of leave alone. For consistency, I use "BAPO" to refer to the entitlement under both the pre-reform rules and the transitional arrangement.

3 Data and Sample Selection

I use Dutch administrative data from Statistics Netherlands, which link secondary-school teachers to demographic characteristics, labor-market outcomes, and health records through unique individual identifiers. The teacher records come from the Education Executive Agency of the Ministry of Education (DUO) and cover the universe of personnel employed in publicly funded secondary schools between 2006 and 2024. Dutch secondary schools are predominantly publicly funded, with privately funded schools accounting for only a small share of the sector.[10] Although the registers also cover management and support staff, I restrict the main analysis to teaching staff. The registers record information on each employment relationship as observed in October of each year, including contract type, salary scale, gross monthly earnings, and annual full-time equivalents (FTE), and, from 2016 onward, the amount of leave taken under BAPO. Contract size defines the two groups analysed throughout: following the Statistics Netherlands definition, I classify teachers as full-time if their 2013 contract is at least 0.95 FTE and as part-time otherwise. I complement the school-reported registers with payroll-based employer–employee records, which identify participation in any subsidized leave program involving a salary reduction and provide annual analogues of the October snapshot variables. Finally, I observe each teacher's gender, immigrant background, and exact day of birth.

I further link teachers to claims data from the Dutch mandatory health insurance, which covers all residents. The claims include reimbursed prescription medications dis-

[9] Pro rata to the contract and with pension accrual unchanged. Teachers in the treatment group could forgo BAPO leave and receive the *basisbudget* instead.

[10] Schools are legally required to report educational personnel to DUO annually. Coverage is near complete: between 2009 and 2020, an average of 2 percent of secondary-school teachers went unreported.

pensed outside hospitals and total healthcare expenditure. Health is a natural outcome of the program, whose purpose is to make teaching sustainable at older ages by easing late-career workload. The strain it targets has both a psychological and a physical component: teachers frequently report psychological complaints (OECD, 2020; van den Heuvel and de Vroome, 2023), and musculoskeletal problems are among the most common physical complaints in the workforce (TNO, 2024). I therefore use antidepressant and pain-medication prescriptions as measures of mental and physical health, respectively. Because the claims cover all residents, health is observed regardless of subsequent employment, for teachers who leave teaching as well as for those who stay.

The main sample consists of teachers employed in secondary education in 2013 for whom teaching was the main source of income in that year, and born between December 1961 and March 1963.[11] Teachers born between December 1961 and July 1962 qualified for the transitional BAPO arrangement and form the treatment group; those born between August 1962 and March 1963 became eligible only at age 57 and form the control group. The resulting sample contains 2,278 teachers, whom I follow from 2015 to 2018 in a balanced panel of 9,112 person-year observations.

I impose two further restrictions. First, I drop 43 teachers who died or emigrated during the panel, whose health outcomes are unobserved; eligibility does not affect the probability of ceasing to be a Dutch resident (Table A.1), so the restriction is unlikely to induce differential selection at the cutoff. This restriction leaves a balanced panel of 2,235 teachers and 8,940 person-years. Second, 2.2 percent of person-years appear in the employer–employee data under an education-sector employer but are absent from the teacher register in the same year, typically because of delayed or erroneous school reporting that the October snapshot misses. Including these observations would require assuming continued employment as a teacher, which might not hold if the individual moved into a support role within the same school. I therefore exclude these observations from the main specification. The final estimation panel is therefore unbalanced, with 8,747 person-years for the same 2,235 teachers.[12] Table A.8 shows that the results are unchanged when these person-years are instead imputed as teaching employment and the panel is kept balanced.

[11] This corresponds to a symmetric eight-month (240-day) bandwidth around the 1 August cutoff. I fix a common bandwidth across outcomes, set to the median of the outcome-specific MSE-optimal bandwidths of Calonico et al. (2014), so that all estimates are identified from the same set of teachers.

[12] Appendix B describes the sample construction in detail.

3.1 Variable Construction

Labor market outcomes come from two administrative sources. I measure *BAPO take-up* using employer–employee records, which report annual participation in any subsidized leave arrangement involving a salary reduction. Given the age composition of the sample, this measure almost exclusively captures BAPO.[13] The remaining outcomes come from the teacher register. *Retention* is an indicator equal to one if an individual is observed as teaching staff in a given year and zero if she is absent from the register or observed in a non-teaching occupation. Moves to temporary agency or self-employed work are classified as exits. From the perspective of school boards, these arrangements represent exits from direct employment: agencies are used primarily when vacancies cannot be filled directly, at higher cost and typically for short spells. In practice, this channel is limited, accounting for 2.3 percent of personnel costs in secondary education in 2013 and 3.7 percent in 2015 (Leemans et al., 2017).

From 2016 onward, I also observe the FTE share allocated to BAPO leave and construct the *leave share*, defined as leave FTE divided by contractual FTE, for teachers who remain in the teacher register. This is the fraction of a teacher's contract taken as subsidized leave; under the collective agreement it cannot exceed 10 percent. Finally, *total teaching time* measures actual teaching hours in FTE, with teachers who have left the register coded as zero. This outcome captures the program's net effect on teaching supplied, combining reductions in hours through leave with changes in retention.

I measure health along several dimensions. *Antidepressants* is an indicator of mental health that equals one if an individual fills any reimbursed prescription for an antidepressant (ATC code N06A) in a given year and zero otherwise (Huber et al., 2013). Because antidepressants can be prescribed in primary care without a formal psychiatric diagnosis, this measure can capture psychological distress that may not appear in diagnosed mental healthcare. I therefore also construct a broader measure of mental health that equals one if an individual either fills an antidepressant prescription or has any reimbursed expenditure on generalist basic mental healthcare.[14]

Pain medication is an indicator of physical health that equals one if an individual

[13] The measure may also include generic unpaid leave classified as "other" (*overig verlof*), which is distinct from maternity and sick leave.

[14] Dutch mental healthcare is organized in echelons. Mild complaints are treated within general practice by a mental health practice nurse, whose services are financed within the GP segment and are not separately identified in the claims data. Patients with more serious complaints are referred either to generalist basic care, for mild to moderate disorders, or to specialist care, for severe or complex disorders. I exclude specialist care because generalist basic care captures the margin of psychological distress that a reduction in workload could plausibly affect.

fills any reimbursed prescription for analgesics or anti-inflammatory drugs in a given year (ATC codes N02A, N02B, or M01A).[15] Because most M01A drugs are available over the counter in the Netherlands, the measure captures pain that is sufficiently severe to generate a reimbursed prescription rather than everyday discomfort.[16]

Finally, I use annual total healthcare expenditure, deflated to 2024 prices, as a summary measure of healthcare use. Expenditure is highly skewed, and its mean is driven by hospital episodes largely unrelated to workload. Because nearly all teachers have some claims each year, an indicator for any expenditure would also carry little information. I therefore code an indicator for expenditure above the median of the 2013 distribution in the analysis sample, which separates years of routine care from years with additional treatment, the margin at which work-related illness is most likely to appear.

Table 1 presents descriptive statistics for the analysis sample in 2013, the year before the reform, for the full sample and separately for full- and part-time teachers. Teachers are 51 years old on average, and 16 percent have an immigrant background.[17] The two groups differ substantially in gender composition: women account for 31 percent of full-time teachers, compared with 73 percent of part-time teachers. Full-time teachers hold contracts of 1 FTE and earn average gross monthly salaries of 4,273 euros, compared with 0.70 FTE and 2,850 euros among part-time teachers. Antidepressant use is twice as high among part-time teachers (8 versus 4 percent), consistent with their higher share of women. Pain-medication use and annual healthcare expenditure are similar across the two groups.

4 Empirical Design

This paper studies the effects of eligibility for subsidized late-career workload reduction on teachers' outcomes. For identification, I exploit the 2014 BAPO reform in a regression discontinuity (RD) design that compares teachers born on either side of an eligibility cutoff. Teachers born before 1 August 1962 had turned 52 by the reform date and retained access to BAPO from that age, whereas those born on or after that date were still 51 and, under the post-reform rules, could take up the leave only from age 57. Date of birth thus generates a sharp discontinuity in program access.

[15] Huber et al. (2013) use N02A and N02B only. I additionally include M01A (anti-inflammatory and antirheumatic products), which are commonly prescribed for neck and back pain associated with the physical demands of teaching. Results are robust to the narrower definition.

[16] This restriction implies that the measure does not capture pain treated exclusively with over-the-counter medication.

[17] An individual or at least one parent was born abroad.

I use the panel structure of the data to estimate the following baseline RD specification:

$$Y_{it} = \alpha + \beta D_i + \gamma(R_i - c) + \delta\, D_i(R_i - c) + X_i'\eta + \lambda_t + \varepsilon_{it}, \tag{1}$$

where Y_{it} is the outcome of teacher i in year t, R_i is teacher i's date of birth expressed in days, c is the cutoff (1 August 1962), and $D_i = \mathbf{1}\{R_i < c\}$ indicates eligibility for BAPO from age 52. The interaction $D_i(R_i - c)$ allows the slope to differ on each side of the threshold, X_i is a vector of baseline covariates and λ_t are year fixed effects [18]. Because I observe the same individuals over multiple years, standard errors are clustered at the individual level. The coefficient of interest, β, measures the discontinuity in outcomes at the cutoff and identifies the effect of BAPO access at age 52. The main specification excludes covariates.

The analysis window spans 2015 to 2018. I exclude 2014 because the treatment group is only partially exposed in that year: eligible teachers could enter BAPO under the pre-reform rules upon turning 52, so most had used the program for a few months by the time outcomes are measured in October 2014. The window ends in 2018, the final year before the control group turns 57 and becomes eligible under the post-reform rules. The main specification pools the four post-reform years to gain precision, and I also report year-by-year estimates to trace the dynamics of the response. To assess whether the effects persist once the control group also gains access, I additionally extend the window to 2024, when the treatment group reaches age 62. I estimate equation (1) by local linear regression within a 240-day bandwidth around the cutoff, using a uniform kernel. I hold the 240-day bandwidth fixed across outcomes to improve the comparability of the estimates and to limit concerns that bandwidth selection drives the results. The chosen window is in line with the median MSE-optimal bandwidths (Calonico et al., 2014) computed separately for the main labor-market and health outcomes. I assess sensitivity to this choice in Section 8.

Heterogeneity by Contract Size The program should not affect all teachers equally. Dutch teaching contracts can be adjusted annually, and many teachers reduce their hours without a subsidy; the average contract was 0.83 FTE in 2013.[19] Such reductions are costly in several ways, however. The teacher forgoes the full wage and permanently lowers

[18] The covariates are measured in 2013 and include gender, immigrant background, FTE, and antidepressant and pain-medication use.

[19] Among teachers born in 1960, 33 percent reduced their contractual hours at least once between ages 46 and 60 (author's calculation from the DUO personnel register).

pension accrual on the hours given up; the school may refuse the request on scheduling or reallocation grounds;[20] and hours once given up are difficult to regain. Moreover, contracts cluster at whole working days (37 percent of teachers are at 1.0 FTE and 15 percent at 0.8, with smaller clusters at 0.7, 0.6 and 0.5), so a reduction from full-time typically means giving up a day, twice what BAPO provides. BAPO leave removes all of these costs at once: it is an annual entitlement that employers cannot refuse, taken within the existing contract, with salary-linked rights calculated on the unreduced salary (VO-raad, 2014). Which of the costs binds differs across teachers, and the design cannot separate them.

For a teacher who would prefer fewer hours but for whom a contract cut is not worth its cost, or is not granted, the choice in practice is between full hours and leaving teaching. Eligibility adds a third option, a reduction without the costs above, and may keep her in teaching. For a teacher already at the hours she prefers at the unsubsidized price, the leave is still worth taking because it is cheaper, but her job already has the value a freely chosen schedule gives it, so she would have stayed anyway; the program changes how much she works rather than whether she stays. Contract size helps separate the two types. A smaller contract reveals a teacher who has at some point reduced her hours despite the costs, or was hired on them, and is therefore more likely to be at her preferred hours. A bigger contract pools teachers who want their hours with teachers who would prefer fewer but have not cut them.

I therefore report all estimates separately for full-time and part-time teachers, classified by their 2013 contract, before the reform was announced, using the Statistics Netherlands threshold of ≥ 0.95 FTE.[21] The retention response should be concentrated among full-time teachers, while among part-time teachers the program should reduce hours without raising retention. Two further results are consistent with this. Ineligible full-time teachers cut their contracts at their own expense, exactly the costly reduction the program replaces (Appendix D). The full-time/part-time gap reappears within the lowest salary scale, where pay is held roughly fixed (Section 7).

Continuity assumption Identification rests on the standard RD continuity assumption (Cattaneo et al., 2020a): in the absence of the reform, potential outcomes would vary smoothly with date of birth at the cutoff. A discontinuity at 1 August 1962 can therefore be attributed to the reform. The assumption could fail if date of birth were systemati-

[20] *Wet aanpassing arbeidsduur*, art. 2.

[21] Results are robust to alternative thresholds (Table A.5).

cally related to unobserved determinants of late-career labor-market or health outcomes, for example through season-of-birth effects on early-childhood health and human capital (Buckles and Hungerman, 2013).

I assess this assumption in two ways. First, I estimate equation (1) using pre-reform values of the main labor-market and health outcomes, separately for the full sample and by contract type. Because these outcomes are measured before teachers become eligible for BAPO, a discontinuity at the cutoff would indicate that birth date predicts outcomes for reasons unrelated to the reform.[22] I find no statistically significant discontinuity for any outcome (Table A.2), and the corresponding RD plots show no visible jump at the cutoff (Figure A.3). Second, I test for discontinuities in predetermined characteristics, including gender and immigrant background. These characteristics are balanced at the cutoff for both full-time and part-time teachers (Table A.2; Figure A.2). The largest discrepancy is a 6 percentage point higher share of women to the left of the cutoff in the pooled sample, which is not statistically significant. Together, these results support the continuity assumption.

No sorting around the cutoff A valid RD design also requires that individuals cannot manipulate the cutoff. This concern is negligible in this setting because the running variable was fixed decades before the reform, and the cutoff was introduced retroactively in the 2014 collective agreement. Consistent with this, Figure A.1 shows a smooth density of the running variable with no evidence of bunching around the threshold. A manipulation test following Cattaneo et al. (2020b) yields no evidence of sorting at the cutoff ($p = 0.204$).

No anticipatory effects or confounding policies. Although the running variable cannot be manipulated, the composition of the sample could still respond to the reform if teachers adjusted their employment in anticipation of losing access to BAPO. I therefore restrict the sample to teachers employed in secondary education in 2013, before the reform was announced, so that the sample is determined prior to any behavioural response.[23]

[22] Teachers born in December 1961 turned 52 during 2013 and were in principle entitled to BAPO. Eligibility begins in the month following the 52nd birthday, however, so no teacher within the estimation bandwidth could take up the program in 2013.

[23] The reform was first set out in the CAO-VO 2014–2015 Negotiators' Agreement, concluded on 15 April 2014, which abolished the BAPO scheme and introduced a new seniority arrangement. While discussions about reform had taken place prior to this date, the exact rules (including the increase in the eligibility age from 52 to 57 and the transitional arrangement based on the 31 July 2014 reference date that effectively determined the cutoff) were only specified in the April agreement.

Finally, the estimates would be confounded if another policy affecting my sample used 1 August 1962 as an eligibility cutoff. I am not aware of any such Dutch policy.

5 Results

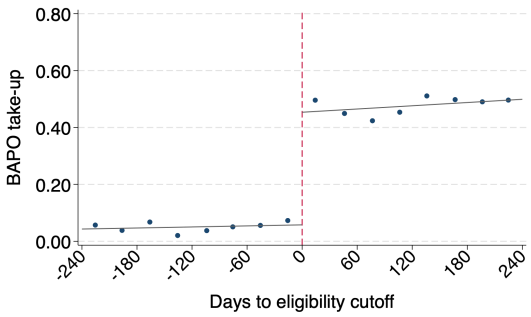
This section reports estimates of equation (1) for the full sample and separately for full- and part-time teachers. RD plots show mean outcomes in 30-day birth-cohort bins, with linear fits on either side of the cutoff. I first examine take-up, then retention, and finally the program's net effect on teaching supply. The main analysis covers 2015-2018, when eligibility differs across groups. I then extend the analysis through 2024, when both groups are eligible.

5.1 Take-up

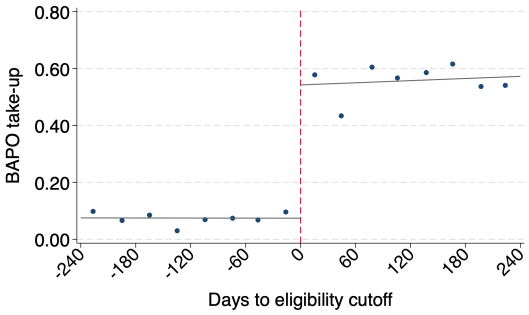
Eligibility increases BAPO take-up by 40 percentage points in the full sample (Table 2; Figure 3), with larger effects for full-time than part-time teachers (47 versus 34 percentage points). Figure A.4 reports year-specific RD estimates for 2010–2020. The gap emerges in 2014 and narrows in 2019, when the control group becomes eligible at age 57, and it does not close fully, consistent with gradual take-up and with deferral reducing the pool of potential participants through exits or unsubsidized reductions in hours.

From 2016 onward, I observe the leave share, defined as subsidized leave FTE relative to contractual FTE. Eligibility increases this share by 0.039 overall, 0.047 among full-time, and 0.033 among part-time teachers. Scaling these estimates by the corresponding take-up effects yields complier leave shares of 0.10 and 0.095, respectively, close to the 0.10 entitlement cap. Participants in both groups therefore use nearly their full entitlement. Because leave share is observed only among teachers who remain in teaching, retention responses may affect the composition of the sample. Section D shows that this selection has only a modest effect on the estimated leave share.

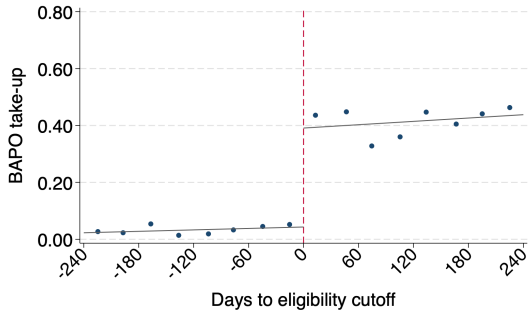
Figure 3: Effect of eligibility on BAPO take-up



(a) All teachers



(b) Full-time



(c) Part-time

Notes: Each panel plots BAPO take-up against date of birth relative to the 1 August 1962 cutoff, pooled over 2015–2018, with evenly spaced 30-day bins and separate linear fits on each side; the bandwidth is 240 days with a uniform kernel. Full-time is defined as ≥ 0.95 FTE in 2013; part-time below.

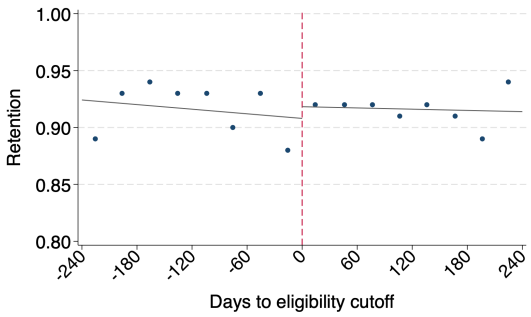
5.2 Retention

Table 3 and Figure 4 report the effect of eligibility on retention in teaching. In the full sample, the estimate is 1.0 percentage point and not distinguishable from zero. This pooled estimate, however, masks a sharp difference by contract type. Among full-time teachers, eligibility increases retention by 6.1 percentage points, significant at the 5 percent level, relative to a control mean of 90 percent. Consistent with this estimate, the RD plot shows a discrete jump at the cutoff against a declining gradient on either side. Among part-time teachers, by contrast, the estimated effect is -2.7 percentage points and statistically insignificant. The difference between the two estimates, 8.8 percentage

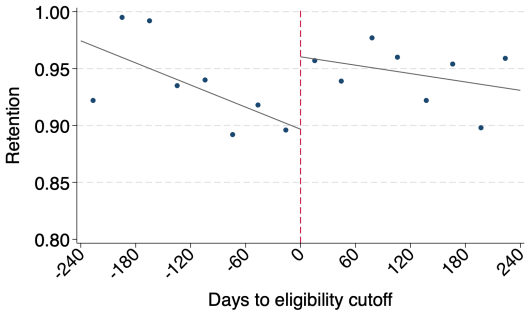
points, is itself significant at the 5 percent level. Taken together, these results suggest that the subsidy primarily retains teachers for whom it relaxes a constraint on hours: the response is concentrated among full-time teachers, rather than among those who work shorter hours.

Year-by-year estimates show that the full-time effect appears in the first post-reform year and persists through 2018, whereas the part-time estimates are insignificant throughout (Figure 5). The gap does not close once the control group becomes eligible at 57: pooling 2015–2024, by which point the treatment cohort has reached 62, the full-time estimate is 6.3 percentage points, significant at the 10 percent level and the part-time estimate is −3.7 points and insignificant. Teachers lost during the five-year deferral are therefore not recovered later. The full-time response reflects durable rather than temporary exits: treating a teacher as having left only if she never reappears in the register yields an estimate of 0.060, equal to the main one (Table A.4), so the retention effect consists of teachers who would otherwise have left for good. For part-time teachers the permanent-exit estimate is −0.049, significant at the 10 percent level and somewhat more negative than the main −0.027, although the two are not statistically distinguishable.

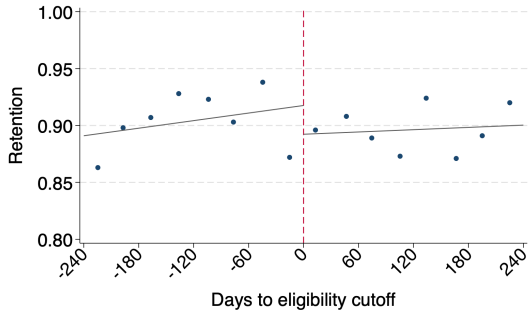
Figure 4: Effect of eligibility on retention in teaching



(a) All teachers



(b) Full-time



(c) Part-time

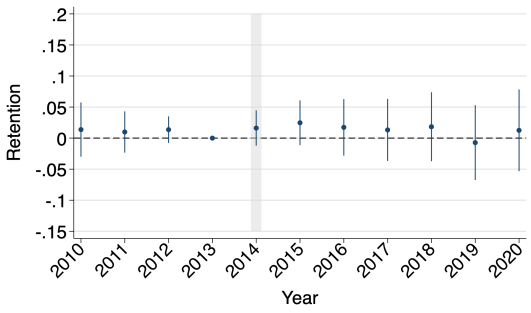
Notes: Each panel plots retention in teaching against date of birth relative to the 1 August 1962 cutoff, pooled over 2015–2018, with evenly spaced 30-day bins and separate linear fits on each side; the bandwidth is 240 days with a uniform kernel. The RD estimate and its standard error, clustered at the teacher level, are reported in each panel. Full-time is defined as ≥ 0.95 FTE in 2013; part-time below.

Columns (2)–(4) of Table 3 decompose the full-time retention effect by destination. Eligibility reduces the probability of holding a job outside education by 2.3 percentage points and raises the probability of being employed in any sector by 2.5 percentage points, both significant at the 10 percent level; the effect on moves into support-staff positions is small and insignificant. The destination estimates are imprecise, however, and the split should be read as indicative.

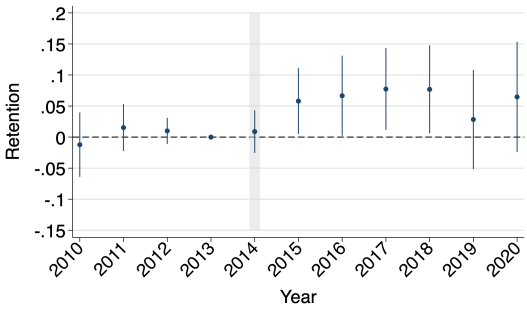
Because take-up is incomplete, the estimates are intent-to-treat effects (ITT). Dividing the full-time estimate by the first stage of 0.47 gives an implied effect of about 13 percentage points among compliers. Reading this ratio as a local average treatment effect (LATE) requires that eligibility affect retention only through program use. That restric-

tion may fail if eligibility is valuable in itself, for instance because the option to reduce hours later keeps a teacher in the job even if she never exercises it, in which case the scaled estimate overstates the effect of participation.

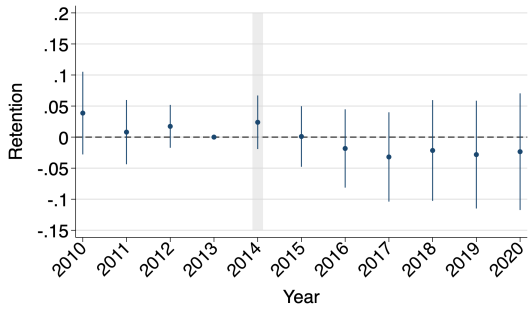
Figure 5: Year-by-year effect of eligibility on retention in teaching, 2010–2020



(a) All teachers



(b) Full-time



(c) Part-time

Notes: Each panel plots year-by-year RD estimates of equation (1) with 95 percent confidence intervals based on standard errors clustered at the teacher level. The shaded band marks 2014, excluded because the treatment group is only partially exposed in that year. Full-time is defined as ≥ 0.95 FTE in 2013; part-time below.

Where the response occurs Because the Statistics Netherlands definition classifies teachers with contracts below 0.95 FTE as part-time, this group may combine teachers with substantially different levels of attachment to teaching. Table A.5 therefore examines heterogeneity by contract size, using mutually exclusive bins and progressively lower thresholds for full-time status. The estimates differ across contract-size groups: 1.4 percentage points for contracts of 0.71–0.94 FTE, compared with −6.8 percentage points

for contracts of 0.70 FTE or less. The negative estimate for teachers with the smallest contracts is consistent with the possibility that eligibility facilitates exits among teachers who are already weakly attached to teaching. For such teachers, the subsidy may relax the financial constraint associated with reducing their teaching hours or leaving teaching altogether, potentially facilitating transitions into self-employment, other jobs, or out of the labor force (Börsch-Supan et al., 2018). The estimate is too imprecise, however, to draw firm conclusions about this mechanism.

Differences in gender composition are unlikely to explain the pattern. Although women make up 73 percent of part-time teachers but only 31 percent of full-time teachers, the full-time effect is similar for men and women (Table A.6).

5.3 Total teaching supplied

Eligibility affects teaching supply through two opposing margins. Teachers who use the leave teach fewer hours, while among full-time teachers, eligibility increases the probability of remaining in teaching.[24] Whether the program ultimately increases or reduces teaching supply therefore depends on which response dominates. To assess this, I estimate the effect on total teaching supplied, coding teachers who are no longer in a teaching position as supplying zero hours. Because leave hours are recorded only from 2016 onward, the analysis covers 2016–2018.

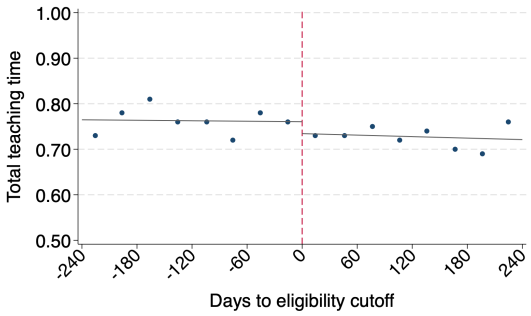
In the full sample, the estimated effect is −0.032 FTE and statistically insignificant (Table 4; Figure 6). The confidence interval rules out an increase larger than 0.015 FTE, equivalent to 2 percent of the control mean. Thus, there is no evidence that the program increases teaching time in the population as a whole.

The pooled estimate again masks two distinct responses. Among part-time teachers, eligibility reduces teaching supplied by 0.067 FTE, a statistically significant effect at the 5 percent level. Because there is no corresponding retention response, the reduction in hours translates directly into lower teaching supplied. Among full-time teachers, by contrast, the estimated effect is 0.040 FTE and statistically insignificant; the confidence interval rules out a net reduction larger than 0.02 FTE, or less than 3 percent of the control mean. For full-time teachers, therefore, the additional hours supplied by those retained in teaching are sufficient to offset the hours released by the leave. Figure 7 reports the dynamic estimates for the full sample and separately for full-time and part-time teachers. The estimates are consistent with an increase in teaching supplied among full-

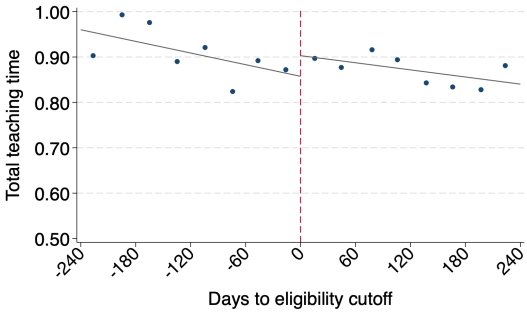
[24] Appendix D discusses hours conditional on remaining in teaching. Because eligibility affects who remains, these conditional estimates are subject to selection and I treat them as descriptive.

time teachers, although the effects are not statistically significant, while teaching supplied declines among part-time teachers following access to the program. Both patterns persist after the control group becomes eligible: pooling 2016–2024, the full-time estimate is 0.056 FTE and insignificant, and the part-time estimate is −0.064 FTE, significant at the 5 percent level (Table 8). The hours lost among part-time teachers are therefore not recovered once both groups have access to the program.

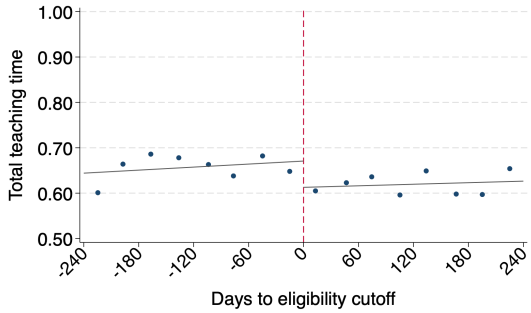
Figure 6: Effect of eligibility on total teaching time



(a) All teachers



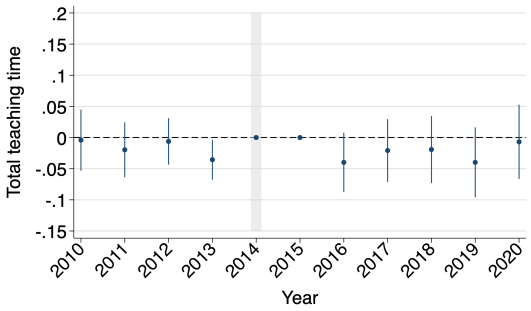
(b) Full-time



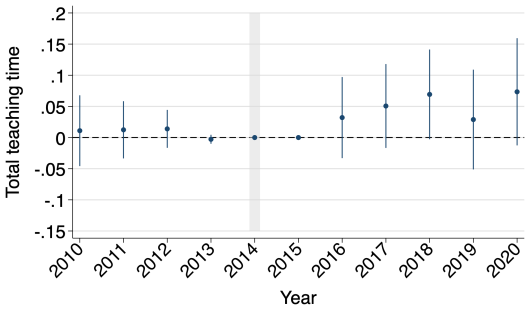
(c) Part-time

Notes: Each panel plots total teaching time against date of birth relative to the 1 August 1962 cutoff, pooled over 2016–2018, with evenly spaced 30-day bins and separate linear fits on each side; the bandwidth is 240 days with a uniform kernel. The RD estimate and its standard error, clustered at the teacher level, are reported in each panel. Full-time is defined as ≥ 0.95 FTE in 2013; part-time below.

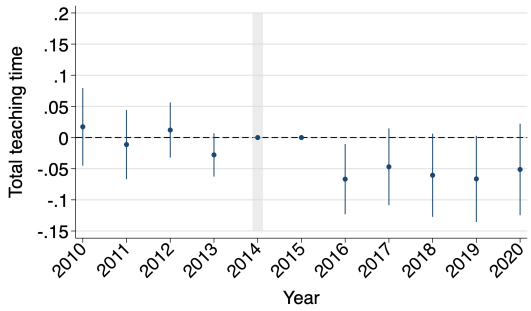
Figure 7: Year-by-year effect of eligibility on total teaching time, 2010–2020



(a) All teachers



(b) Full-time



(c) Part-time

Notes: Each panel plots year-by-year RD estimates of equation (1) with 95 percent confidence intervals based on standard errors clustered at the teacher level. Leave hours are recorded only from 2016; before 2016, total teaching time is measured as contractual FTE, with teachers no longer in the teacher register coded as zero. The shaded band marks 2014, excluded because the treatment group is only partially exposed in that year. Full-time is defined as ≥ 0.95 FTE in 2013; part-time below.

6 Health Outcomes

Eligibility affects both hours worked and retention in teaching, with potentially offsetting implications for health. Among part-time teachers, eligibility reduces hours without affecting retention, isolating the effect of a lighter workload. Among full-time teachers, it reduces hours but also increases retention, making the net change in exposure to a demanding occupation ambiguous. Health outcomes are observed for all teachers in the 2013 sample regardless of subsequent employment, so these estimates are not subject to

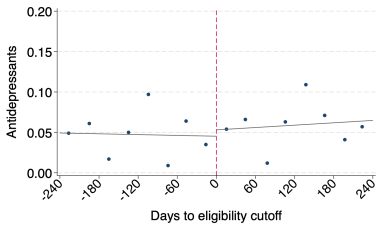
selection from leaving teaching. Table 6 reports the estimates, and Figure 8 shows the corresponding RD plots.

In the full sample, eligibility reduces pain-medication use by 5.7 percentage points, relative to a control mean of 21 percent, significant at the 1 percent level, but has no detectable effect on antidepressant use or high healthcare expenditure. The antidepressant estimate remains insignificant when mental health is measured more broadly as any use of generalist basic mental healthcare (Table A.7). There is also an increase of 5.2 percentage points in the probability of high general-practitioner expenditure, which is consistent with eligible teachers having more time to seek routine care rather than with worse health, given the decline in pain medication and the absence of any change in total expenditure.

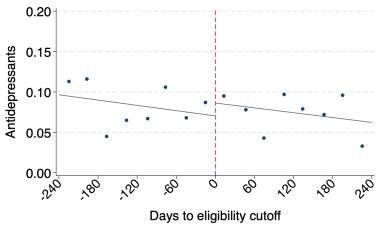
Among part-time teachers, eligibility reduces pain-medication use by 6.6 percentage points, significant at the 5 percent level, with no detectable effect on antidepressant use or high healthcare expenditure. Because this group's retention does not show any significant change, the estimate more closely reflects the effect of a lighter workload and indicates improved physical health.

Among full-time teachers, none of the three outcomes changes significantly. The estimate for pain-medication use is -4.1 percentage points, similar in magnitude to the part-time estimate but imprecise. The confidence intervals rule out increases larger than 2.8 percentage points in pain-medication use, 4.6 in antidepressant use, and 9.1 in high healthcare expenditure. The estimates therefore exclude large increases in pain-medication use but not moderate deteriorations in mental health or healthcare use. The longer careers induced among full-time teachers are not accompanied by detectable harm on the outcomes examined, but the data cannot establish whether the additional years offset the health gain from fewer hours.

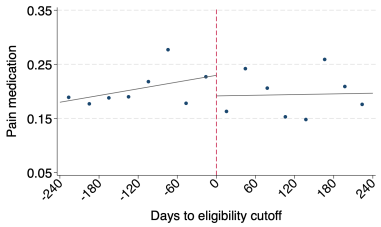
Figure 8: RD plots for health outcomes, by contract type



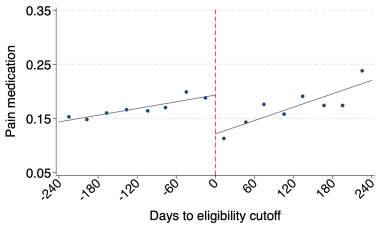
(a) Full-time: antidepressants



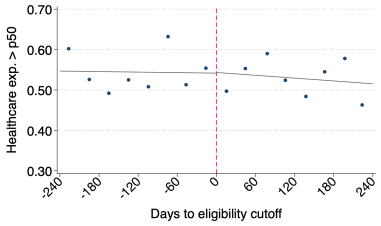
(b) Part-time: antidepressants



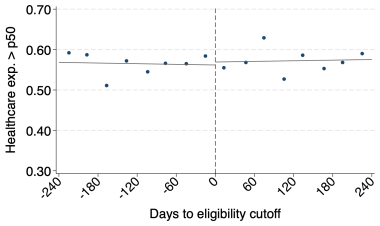
(c) Full-time: pain medication



(d) Part-time: pain medication



(e) Full-time: healthcare expenditure (>p50)



(f) Part-time: healthcare expenditure (>p50)

Notes: Each panel plots the health outcome indicated in the panel title against date of birth relative to the 1 August 1962 cutoff, pooled over 2015–2018, with evenly spaced 30-day bins and separate linear fits on each side; the bandwidth is 240 days with a uniform kernel. Outcomes are observed for all teachers in the 2013 sample and are therefore not conditional on remaining in teaching. Full-time is defined as ≥ 0.95 FTE in 2013; part-time below.

7 Mechanisms

Eligibility lowers the price of working fewer hours, and this can raise retention for two reasons. Fewer hours may relieve strain enough to make a demanding job sustainable, or cheap time off may be valued in itself, regardless of strain. To shed some light on these channels, I compare workers who are offered the same leave program but face different levels of strain. If leave retains workers because cheaper leisure is attractive in itself, both groups should stay longer; if it retains them by making a demanding job sustainable, only

the high-strain group should respond.

I compare full-time teachers in the lowest salary scale (LB) with full-time administrative and educational-support staff. Restricting teachers to the lowest scale brings their pay close to that of support staff, so the comparison holds financial conditions roughly constant and varies strain. Both groups face the same eligibility rules, with a slightly lower co-payment for support staff (40 rather than 50 percent of forgone salary). In the linked survey sample (Table C.2), the two groups have similar contract sizes, gender composition and baseline health, even if LB teachers earn about a fifth more. They differ sharply in strain. Support staff report less work pressure (37 versus 51 percent) and far less emotionally demanding work (10 versus 33 percent), and fewer name shorter hours as a condition for working longer (49 versus 67 percent). They report more physically demanding work (24 versus 14 percent), which plausibly reflects technical, facilities and caretaking roles.

Both groups use the program: eligibility raises take-up by 39 percentage points among LB teachers and by 33 among support staff (Table 10), but only teachers stay longer. Retention among LB teachers rises by 15.4 percentage points, significant at the 1 percent level, against a control mean of 83 percent; the estimate for support staff is 0.7 percentage points and insignificant.[25] The LB effect is also larger than the 6.1 points found for full-time teachers as a whole, and the group differs in ways that might indicate higher strain: LB teachers have a higher exit rate (17 percent against 10 in the control group), are concentrated in the pre-vocational track, where 61 percent of teachers hold the lowest scale against 22 percent in the pre-university track, and hold positions without the coordinating duties that take higher-scale teachers partly out of the classroom (Bisschop and Heyma, 2016).

The comparison between LB teachers and support staff has a limit. Only 3 percent of support staff in the control group leave, so no effect larger than 3 percentage points is possible for them, and the confidence interval, from -3.7 to 5.1, does not exclude that eligibility prevents all of their exits. The null therefore cannot show whether cheap leave would retain low-strain workers if more of them were at risk of leaving. What the comparison does show is that demand for the leave does not depend on strain, while retention responds only where strain and exit risk are both high, consistent with the program keeping teachers by relieving strain rather than by offering cheap leisure.

[25] Support staff pay less for leave, so the subsidy is a slightly larger share of their salary than it is for teachers. If the program worked through the value of the leave itself, they should respond at least as much; weaker financial incentives cannot explain their null.

8 Robustness

Bandwidth sensitivity Figures A.5–A.7 report estimates of equation (1) across bandwidths ranging from 90 to 330 days. Take-up remains stable across bandwidths in both groups. For full-time teachers, the retention effect loses statistical significance only at the narrowest bandwidths, while the point estimate remains essentially unchanged, consistent with the loss of precision from the smaller sample rather than instability in the estimate. For part-time teachers, the retention estimate moves toward zero and becomes more variable at narrow bandwidths. The estimate for total teaching time among full-time teachers declines somewhat at narrower bandwidths but remains statistically insignificant throughout. Among part-time teachers, the estimate likewise loses statistical significance at narrow bandwidths while remaining similar in magnitude. Health estimates are generally stable across bandwidths, with the exception of pain medication use and health expenditure among part-time teachers, for which the estimates tend to become smaller at wider bandwidths.

Placebo cutoffs As a falsification test, I re-estimate equation (1) at 25 placebo cutoffs spaced one month apart over birth dates from May 1963 to May 1965, at which eligibility does not change (Cattaneo et al., 2020b).[26] The nine-month gap after 1 August 1962 ensures that no placebo window overlaps the true discontinuity. Ranking the true estimate among the 26 estimates (true plus placebos) in the direction of its sign, from the largest for positive estimates and from the smallest for negative ones, yields a one-sided randomization-inference p-value equal to the rank divided by 26 (Ganong and Jäger, 2018). Table A.13 reports the results.

Take-up ranks first in all three samples, as do retention among full-time teachers and teaching supplied among part-time teachers ($p = 1/26 \approx 0.04$, the smallest the exercise can deliver): at no other birth date do the data produce a jump as large. Among the health outcomes, the decline in pain-medication use ranks first in the full sample, third among part-time teachers and fifth among full-time teachers, while antidepressant use and healthcare expenditure rank low in every sample, consistent with their null conventional estimates.

Balanced panel The main specification drops the 193 person-years that appear in the employer–employee data under an education-sector employer but not in the teacher register. Table A.8 instead imputes these as teaching employment, keeping the panel

[26] I use data until 1965 because day of birth is available in the data only through that year.

balanced at 8,940 person-years. Every coefficient lies within a standard error of its main-specification counterpart, so the treatment of these observations does not drive any of the results.

Adding control variables I re-estimate the main specification including the full vector of baseline covariates (Table A.9). Take-up, full-time retention and the health estimates are essentially unchanged. The part-time retention coefficient moves from −0.027 to −0.017 and remains insignificant. The one change is total teaching supplied among part-time teachers, which falls from −0.067 to −0.039 and loses significance, as the 2013 contract might absorb a small, insignificant difference in contract size at the cutoff within this group (Table A.2).

Local randomization Local randomization inference within ±30, ±60 and ±90 days of the cutoff gives the same picture (Table A.3). Take-up jumps in every window for both groups; full-time retention is stable at 0.04–0.06 and significant under randomization inference at ±90 days, while part-time retention is close to zero in every window (Cattaneo et al., 2016).

Other specification checks The main estimates are not driven by the choice of bandwidth, kernel, or functional form. Table A.12 re-estimates every outcome at the MSE-optimal bandwidth selected separately for each outcome and sample, using a triangular kernel and robust bias-corrected inference (Calonico et al., 2014). The point estimates are close to those of the main specification and their significance is unchanged, with one exception: total teaching supplied among full-time teachers becomes significant at the 10 percent level, consistent with the positive but imprecise main estimate. Allowing a second-order polynomial in the running variable on each side of the cutoff (Table A.10) widens the confidence intervals, as expected, but leaves the pattern intact: take-up remains large and precisely estimated, full-time retention is 0.083 and significant at the 10 percent level, and the pain-medication estimates keep their sign and magnitude. The part-time retention coefficient changes sign to 0.030 and insignificant. A donut specification drops observations within one month of the cutoff (Table A.11). For full-time teachers, the coefficients are essentially unchanged, with retention at 0.058 against 0.061 in the main specification, though retention loses significance with the smaller sample. For part-time teachers, pain medication and teaching supplied remain negative and of similar magnitude but lose significance, while retention becomes more negative (−0.059, significant at the 10 percent level). The check is demanding, as it excludes the teachers closest

to the cutoff and hence the most informative, which accounts for the reduced precision.

No disappointment response in the control group Teachers born just after the cutoff learn that a few months of birth date cost them five years of eligibility. A narrow loss of this kind can itself trigger a response, in the form of exit or worse mental health, that would confound the RD comparison (De Grip et al., 2012). It is reasonable to expect that any such response is strongest for the cohort that missed most narrowly and fades with distance from the cutoff, since later cohorts were excluded under the same rules but were still several years away from turning 52 when the reform was announced. I therefore compare the RD control cohort (born August 1962 to March 1963) with cohorts born one and two years later, in a difference-in-differences design with the most distant cohort as reference.[27] If there is a disappointment effect, I would expect the Cohort 1 interaction to be negative for retention and positive for medication use, and larger in absolute value than the Cohort 2 interaction. Because the cohorts are a year apart in age, the interactions might also absorb differences in the age profile of the outcomes; this exercise is therefore suggestive evidence rather than a test.

Table A.14 shows no such response. Among full-time teachers, where the retention effect is found, the Cohort 1 interaction is -0.010, insignificant, and no larger than for Cohort 2, so the narrowly-missing cohort left no more often than cohorts further from the cutoff. Among part-time teachers the interactions are -0.026 and -0.016; the decline across cohorts is roughly uniform per year of age, with no additional drop for Cohort 1. A disappointment response in this group would in any case bias the part-time estimate upward and so could not produce a spurious retention gain. Antidepressant use, the margin on which De Grip et al. (2012) find disappointment to operate, is flat in every panel, and the pain-medication interactions are negative or close to zero, except among full-time teachers, where the Cohort 1 coefficient is positive but insignificant.

9 Spillovers

The individual-level estimates establish what eligibility does to a teacher's own hours, but not how schools absorb it. For full-time seniors, eligibility leaves total teaching supplied unchanged while changing its form: a retained teacher on leave opens a fractional gap in the timetable, whereas the exit she would otherwise have made opens a full vacancy. For

[27] The sample consists of teachers employed in secondary education in 2013 in the three cohorts, observed over 2011–2018, with cohort and year fixed effects; Post equals one for 2015–2018.

part-time seniors, the leave passes through and capacity is lost outright. Either way the lessons must be covered, by adding hours to incumbent colleagues or by hiring, and which margin schools use determines whether a program that retains experienced teachers also thins the teaching capacity of the schools that employ them. This section exploits cross-school variation in exposure to the transitional arrangement to trace how the withdrawn hours are covered.

Treatment I exploit cross-school variation in the share of senior teachers eligible for BAPO. A higher exposure share implies that more senior teachers are eligible and, therefore, that schools face greater potential withdrawals of teaching hours. Exposure is the 2013 FTE-weighted share of a school's teachers who kept early access under the transitional arrangement, that is, teachers born between August 1958 and July 1962:

$$\pi_s = \frac{\sum_{j \in s} \text{FTE}_{j,2013} \cdot \mathbf{1}[\text{born Aug 1958–Jul 1962}]}{\sum_{j \in s} \text{FTE}_{j,2013}}. \tag{2}$$

I use the full four-year retained cohort rather than birth dates near the 1 August 1962 cut-off alone, because a narrow window would generate little cross-school variation.[28] Because π_s is fixed by the 2013 workforce, it is predetermined with respect to any post-reform adjustment. I restrict the analysis to schools employing at least 15 teachers in 2013. Until August 2014 every teacher aged 52 or above could reduce her hours, so a school with many teachers born in 1958–1962 was in no way distinctive. From August 2014, these are precisely the teachers who retain early access while their marginally younger colleagues are deferred to 57. The difference-in-differences asks whether outcomes in high-π_s schools diverge after 2014 from those in schools whose seniors were deferred.

Identification The exposure share is not, by itself, a measure of the number of hours withdrawn. A school may have a high π_s simply because its workforce is older, and age structure shapes retirements and turnover independently of BAPO. Teachers born before August 1958, moreover, were eligible under both regimes, so a school with few retained teachers may still see substantial BAPO use among its oldest staff. I therefore control directly for the 2013 FTE-weighted shares of the adjacent deferred cohort, π_s^{def} (born August 1962–July 1966), of always-eligible older teachers, π_s^{old} (born before August 1958), and of younger teachers, π_s^{young} (born 1967–1975), each interacted with year indicators.

[28] Teachers born before August 1958 were already 57 by 2015, so under the new rules they would have been eligible in any case.

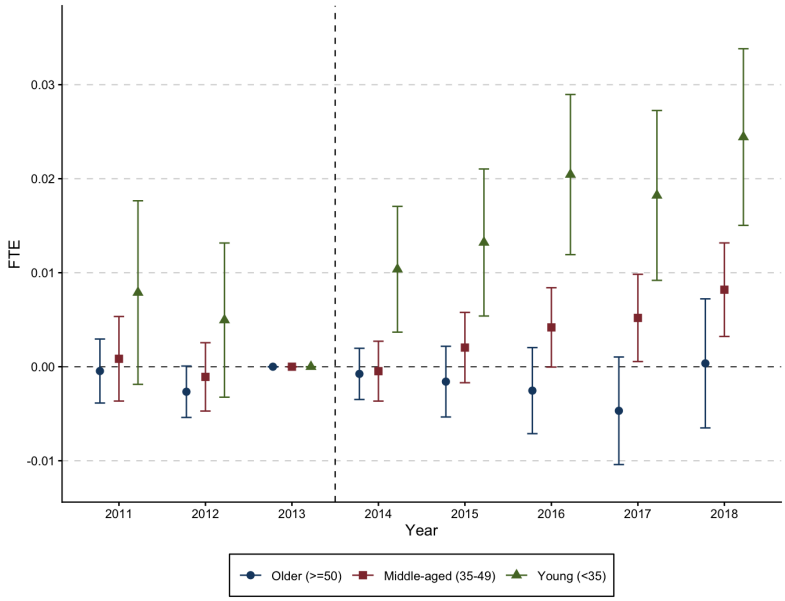
The identifying comparison is thus between schools with similar age structures that differ only in the share of teachers who retained early access.

Incumbent teachers I first ask whether withdrawn hours are absorbed by colleagues. The outcome is the labor supply of incumbents who are themselves untreated throughout the window, born on or after 1 August 1962 and hence ineligible before 2019, so that any response to school exposure reflects their colleagues' treatment rather than their own. I estimate

$$y_{ist} = \alpha_i + \lambda_t + \sum_{t \neq 2013} \beta_t \left(\pi_s \times \mathbf{1}[\text{year} = t]\right) + \sum_{t \neq 2013} \left(\gamma_t \, \pi_s^{\text{def}} + \delta_t \, \pi_s^{\text{old}} + \eta_t \, \pi_s^{\text{young}}\right) \times \mathbf{1}[\text{year} = t] + \varepsilon_{ist}, \tag{3}$$

where y_{ist} is the actual working FTE of teacher i, assigned to her 2013 school s and set to zero if she is no longer employed as a teacher, so that the estimates capture total hours delivered rather than hours conditional on staying. Teacher and year fixed effects are α_i and λ_t, and standard errors are clustered at the individual level. Each β_t measures the differential evolution of untreated incumbents' hours in more exposed schools, relative to 2013. Identification requires that, absent the reform, schools with different exposure would have evolved in parallel; the pre-2014 coefficients provide the corresponding test. Figure 9 reports event-study coefficients separately for young (< 35), middle-aged (35–49), and older untreated (≥ 50) incumbents, with age defined in 2013. Table A.15 reports the corresponding pooled difference-in-differences estimates. The response is concentrated among young incumbents: a 10 percentage point increase in school exposure raises their working FTE by 1.8 percentage points over the post-reform period, with the year-by-year estimates showing a gradual increase. The pooled estimates for middle-aged and older incumbents are 0.2 and 0.02 percentage points, respectively, neither distinguishable from zero. Young teachers also have the most scope to expand, holding contracts further below the full-time norm.

Figure 9: Event-study estimates of school exposure on incumbent teachers' working hours, by age group

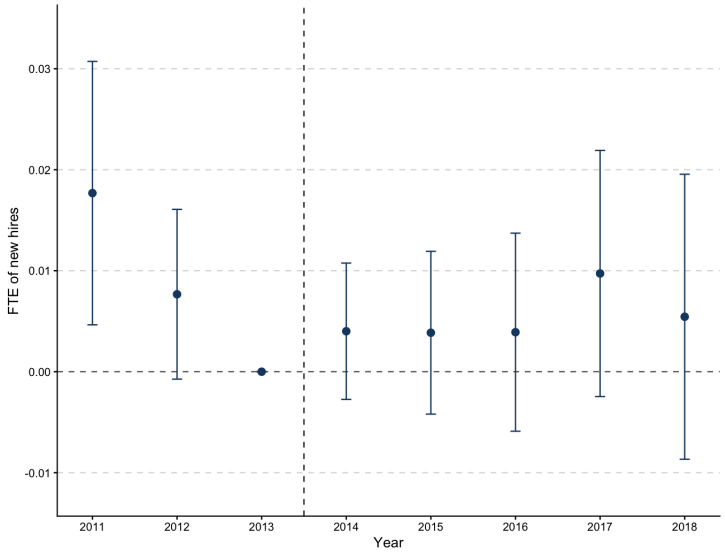


Notes: The figure plots event-study coefficients β_t from equation (3), separately for young (< 35), middle-aged (35–49), and older (≥ 50) incumbents, with age defined in 2013 and 2013 as the omitted reference year. Coefficients are scaled to a 10 percentage point increase in the exposure share π_s. The outcome is actual working FTE, set to zero for teachers no longer employed as teaching staff. The sample is restricted to schools with at least 15 teachers in 2013. Bars denote 95 percent confidence intervals based on standard errors clustered at the individual level.

New hiring A second adjustment margin is hiring. I estimate the same event study at the school level, using the FTE of teachers newly employed at the school in year t, relative to the school's 2013 capacity. Figure 10 shows no response to exposure after 2014. Hiring therefore does not appear to be the margin through which schools adjust. The caveat is that agency and self-employed teachers are not recorded in the register, so the null concerns direct employment only.

Together with the incumbent results, this pattern suggests that schools accommodate the leave of retained seniors by increasing hours among younger teachers rather than hiring additional staff. The program therefore changes who delivers teaching rather than reducing schools' teaching capacity.

Figure 10: Event-study estimates of school exposure on new hiring



Notes: The figure plots event-study coefficients from the school-level analogue of equation (3), with school and year fixed effects and 2013 as the omitted reference year. Coefficients are scaled to a 10 percentage point increase in the exposure share π_s. The outcome is the FTE of teachers employed at the school in year t who were not employed there in year $t-1$, relative to 2013 capacity, constructed from the teacher register. The sample is restricted to schools with at least 15 teachers in 2013. Bars denote 95 percent confidence intervals based on standard errors clustered at the school level.

10 Cost-effectiveness

Among full-time teachers, eligibility raises the leave share by 0.045 FTE per eligible teacher (Table 2).[29] At a 65 percent subsidy and a mean salary of €51,300 for full-timers and €34,200 for part-timers, the program costs about €1,500 per eligible full-time teacher per year and €660 per part-time teacher. Dividing the full-time outlay by the 6.1 percentage-point retention effect gives about €25,000 per retained teacher-year, half a salary.

The benefit of a retained teacher-year is the avoided cost of covering the vacancy. In October 2024, 31 percent of secondary-school shortage hours were open vacancies, 51 percent were covered internally, three quarters of them by a teacher not qualified in the subject, and 17 percent by agency or self-employed staff (Ministerie van Onderwijs, Cultuur en Wetenschap, 2024, Table 11). Table 11 values each mode relative to keeping the senior, in money and in student achievement per year of exposure: 0.05–0.11 standard

[29] 0.047 among those who remain, scaled by 96 percent retention; the 2015 leave share is assumed equal to the 2016–2018 average.

deviations when the lessons are taught out of field (Clotfelter et al., 2010), 0.02–0.06 when taught by a teacher in her first five years (Rivkin et al., 2005; Clotfelter et al., 2010), and zero for an experienced colleague. Weighted, a vacancy-year costs about 0.05 of a salary more than the senior, because agency staff cost nearly twice as much per hour, and about 0.06 standard deviations of achievement, because 77 percent of shortage hours are taught out of field.

I value achievement with Chetty et al. (2014b): a one-year, teacher-induced gain of 0.10 student standard deviations is worth about $7,000 per student in lifetime earnings, so 0.01 standard deviations across the 150 students a full-time teacher teaches is worth about €140,000 in 2024 prices.[30] The avoided vacancy is thus worth about €840,000. The 0.74 FTE of leave is absorbed by incumbents expanding their contracts (Section 9); assuming a third of them are in their first five years, these hours are 0.013 standard deviations less effective than the senior's, a loss of about €130,000. The net benefit per retained full-time teacher-year is about €680,000,[31] and remains positive as long as achievement is worth more than 3 percent of the Chetty value.

At the program level, eligibility also covers the 57 percent of teachers who are part-time. Per 100 eligible teachers per year the outlay is €102,000; retention yields $43 \times 0.061 = 2.6$ full-time teacher-years worth about €2.2 million; and 5.7 FTE of teaching shifts to younger colleagues, saving €44,000 in wages but costing about €1.0 million in achievement.[32] The net is about +€1.1 million: +€1.8 million for full-time and −€0.7 million for part-time teachers. The program pays off as long as the gap on reallocated hours stays below about 0.03 standard deviations, twice the central estimate; this threshold does not depend on the earnings valuation. Restricting eligibility to full-time teachers would cut outlays by 37 percent and remove the component that cannot pay off. The calculation omits reductions in sickness absence, the value teachers place on the leave, and avoided non-employment for the 40 percent of retained teachers who would otherwise have left employment.

For a government filling vacancies, the relevant alternative is the cost of an additional teacher-year through recruitment or other retention policies. The BAPO subsidy costs

[30] $7,000 is the present value at age 12 of a 1.3 percent earnings gain, in 2010 dollars at a 5 percent discount rate, for one standard deviation of teacher value added; one teacher standard deviation equals 0.12 student standard deviations in mathematics and 0.08 in English (Chetty et al., 2014a), and I use the average of 0.10, close to the 0.11 lower bound of Rivkin et al. (2005). The estimates come from US grades 4–8 and project earnings beyond age 28, so their magnitude may not transfer to Dutch secondary schools.

[31] €840,000 + €2,600 (money saving) − €130,000 − €25,000.

[32] $43 \times 0.045 + 57 \times 0.067$ FTE; wages at 0.85 of senior pay; 0.013 standard deviations per FTE-year.

about €25,000 per retained full-time teacher-year. The Dutch lateral-entry subsidy costs €8,000–€25,000 per additional teacher-year under plausible additionality,[33] and English training bursaries about £20,000 (McLean et al., 2023), both for a novice year delivered one to two years later wherever the entrant chooses. Early-career retention payments, the best-evidenced retention alternative, cost about £35,000 per teacher-year retained (Sims and Benhenda, 2022). The BAPO subsidy is thus competitive per teacher-year and buys an experienced year in the school with the vacancy, but it is bounded by the number of teachers near exit and works only for full-timers.

11 Conclusion

Population ageing is pushing governments to extend working lives, yet workers in demanding occupations often leave their profession well before any retirement age binds. This paper asks whether reducing workload late in the career can keep such workers in their jobs. I exploit a 2014 reform of the Dutch BAPO program that deferred eligibility for subsidized leave from age 52 to 57 for teachers born on or after 1 August 1962, and compare teachers on either side of this birth-date cutoff in a regression discontinuity design. Administrative records covering all Dutch secondary-school teachers, linked to health-insurance claims, allow me to follow retention, teaching supply, and health outcomes.

Eligibility raises take-up by 40 percentage points but has no detectable effect on retention or total teaching supply in the full sample. The null conceals two opposing responses. Among full-time teachers, eligibility raises the probability of remaining in teaching by 6.1 percentage points, an effect that appears in the first year and persists through 2024, after the control cohort has itself become eligible. The destination estimates are imprecise but suggest that most retained teachers would otherwise have moved to another job, largely outside education, with the rest leaving salaried employment. Their additional teaching offsets the hours released through leave. Among part-time teachers, eligibility raises take-up but not retention, and teaching supply falls by 0.067 FTE. The subsidy thus retains teachers for whom it relaxes a constraint on hours and finances reductions for teachers who had already adjusted their contracts.

Linked health-insurance claims data show that eligibility reduces pain-medication use by 5.7 percentage points, with no detectable change in antidepressant use or healthcare

[33] €25,000 per entrant (Dienst Uitvoering Onderwijs, 2025) plus similar school costs (Berenschot, 2019), 8–10 discounted teacher-years per entrant, 20–50 percent additionality, which no evaluation has estimated.

expenditure. The reduction is concentrated among part-time teachers, whose retention does not change, so it largely isolates the effect of a lighter workload.

Full-time support staff, who receive the same subsidy but report far less work pressure, take up the leave at similar rates yet show no retention response, whereas full-time teachers in the lowest salary scale, whose pay is closest to theirs, stay 15 percentage points longer. Demand for the leave does not depend on strain, but retention responds only where strain and exit risk are both high, suggesting that the program retaining workers by relieving strain rather than by offering cheap leisure.

These findings inform policies to extend working lives, particularly in demanding occupations. Raising the retirement age does little for workers who leave long before it binds, but changing their working conditions may keep them employed. A subsidized reduction in hours retains workers who want fewer hours but cannot obtain them cheaply, while for those who have already adjusted their hours it reduces teaching without buying retention. As implemented, the program raises neither retention nor total teaching supply overall, but it still yields a positive net return, because the experienced full-time teachers it retains are worth more than the teaching lost among part-time teachers. Restricting eligibility to full-time teachers would preserve the retention gain at a substantially lower cost. So targeted, the subsidy is competitive per retained worker-year with subsidies for lateral entrants, training bursaries, and early-career retention payments, and it delivers an experienced worker immediately, in the school where the vacancy would otherwise arise.

Tables and Figures

Table 1: Descriptive statistics

	Whole sample		Full-time		Part-time	
	Mean	SD	Mean	SD	Mean	SD
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Demographics						
Female	0.55	0.50	0.31	0.46	0.73	0.45
Immigrant background	0.16	0.37	0.16	0.36	0.17	0.37
Age	50.9	0.47	50.9	0.47	50.9	0.48
Panel B: labor market						
Contractual working hours (FTE)	0.83	0.19	1.01	0.03	0.70	0.15
Monthly salary (euros)	3,456	983	4,273	541	2,850	777
Panel C: Health						
Antidepressants	0.06	0.24	0.04	0.21	0.08	0.27
Pain medication	0.18	0.39	0.18	0.38	0.18	0.39
Total healthcare expenditure (euros)	1,715	5,938	1,496	5,265	1,877	6,387
Observations	2,235		952		1,283	

Notes: The table reports means and standard deviations for the analysis sample in 2013, the year before the reform. Labor-market variables come from the DUO teacher register and are measured in October; health variables come from basic health-insurance claims and are measured over the calendar year. Monetary values are in euros, deflated to 2024 prices. The sample consists of teachers employed in Dutch secondary education in 2013 and born within 240 days of the August 1, 1962, cutoff. Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below.

Table 2: Effect of eligibility on BAPO take-up and leave share

	BAPO take-up (1)	Leave share (2)
Panel A: All		
RD estimate	0.395***	0.039***
	(0.031)	(0.003)
Control mean	0.059	0.000
Observations	8,747	5,921
Panel B: Full-time		
RD estimate	0.471***	0.047***
	(0.049)	(0.005)
Control mean	0.078	0.001
Observations	3,740	2,595
Panel C: Part-time		
RD estimate	0.344***	0.033***
	(0.038)	(0.004)
Control mean	0.043	0.001
Observations	5,007	3,326

Notes: The table reports RD estimates of equation (1). BAPO take-up is an indicator for participating in any partially paid leave arrangement at any point during the year, pooled over 2015–2018. Leave share is the ratio of subsidized leave FTE to contractual FTE, pooled over 2016–2018, the period for which leave hours are recorded, and conditional on appearing in the teacher register. Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below. The bandwidth is 240 days around the August 1, 1962, cutoff, with a local linear specification and a uniform kernel. Standard errors, in parentheses, are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 3: Effect of eligibility on retention and destination of exits, by contract type

	Retention	Non-education	Support staff	Total employment
	(1)	(2)	(3)	(4)
Panel A: All				
RD estimate	0.010	−0.015*	0.005	−0.004
	(0.020)	(0.009)	(0.011)	(0.013)
Control mean	0.912	0.026	0.017	0.960
Observations	8,747	8,747	8,747	8,747
Panel B: Full-time				
RD estimate	0.061**	−0.023*	−0.007	0.025*
	(0.028)	(0.012)	(0.020)	(0.013)
Control mean	0.900	0.028	0.019	0.963
Observations	3,740	3,740	3,740	3,740
Panel C: Part-time				
RD estimate	−0.027	−0.009	0.015	−0.024
	(0.028)	(0.012)	(0.012)	(0.021)
Control mean	0.921	0.024	0.007	0.960
Observations	5,007	5,007	5,007	5,007
Panel D: Difference (full-time − part-time)				
RD estimate	0.088**	−0.014	−0.022	0.048*
	(0.040)	(0.017)	(0.023)	(0.025)
Observations	8,747	8,747	8,747	8,747

Notes: The table reports RD estimates of equation (1), pooled over 2015–2018. Retention is an indicator for appearing in the teacher register as teaching staff in October of the year. Columns 2–4 decompose destinations: Non-education is an indicator for holding a job outside the education sector, Support staff for remaining in education in an administrative or educational-support role, and Total employment for being employed in any sector; all are constructed from the teacher and employer–employee registers and measured in October. Panel D reports the difference between the full-time and part-time estimates. Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below. The bandwidth is 240 days around the August 1, 1962, cutoff, with a local linear specification and a uniform kernel. Standard errors, in parentheses, are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 4: Effect of eligibility on total teaching supplied, by contract type

	Unconditional actual working hours (1)
Panel A: All	
RD estimate	−0.032
	(0.024)
Control mean	0.764
Observations	6,524
Panel B: Full-time	
RD estimate	0.040
	(0.032)
Control mean	0.863
Observations	2,787
Panel C: Part-time	
RD estimate	−0.067**
	(0.028)
Control mean	0.680
Observations	3,737

Notes: The table reports RD estimates of equation (1). The outcome is total teaching supplied, that is, contractual FTE net of subsidized leave, with teachers no longer observed in the teacher register coded as zero. The sample is pooled over 2016–2018, the period for which leave hours are recorded. Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below. The bandwidth is 240 days around the August 1, 1962, cutoff, with a local linear specification and a uniform kernel. Standard errors, in parentheses, are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 5: Effect of eligibility on working hours and monthly wage

	Total working hours (1)	Total wage (2)
Panel A: All		
RD estimate	-0.039^{*}	-92.5
	(0.020)	(129.4)
Control mean	0.804	4,676.0
Observations	6,494	8,747
Panel B: Full-time		
RD estimate	0.019	289.9^{**}
	(0.020)	(139.4)
Control mean	0.918	5,454.2
Observations	2,769	3,740
Panel C: Part-time		
RD estimate	-0.060^{**}	-241.8
	(0.026)	(153.6)
Control mean	0.703	3,995.4
Observations	3,725	5,007

Notes: The table reports RD coefficients estimated from equation (1). *Total working hours* is working hours summed over all jobs held, pooled over 2015–2018. *Total wage* is the monthly gross salary in October, in euros. Full-time is defined as ≥ 0.95 FTE in 2013; part-time below. The bandwidth is 240 days around the 1 August 1962 cutoff, with a local linear specification and a uniform kernel. Standard errors in parentheses are clustered at the teacher level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6: Effect of eligibility on antidepressant use, pain-medication use, and healthcare expenditure, by contract type

	Antidepressants	Pain medication	Healthcare expenditure (>p50)
	(1)	(2)	(3)
Panel A: All			
RD estimate	0.007	−0.057***	0.003
	(0.019)	(0.022)	(0.031)
Control mean	0.064	0.212	0.560
Observations	8,747	8,747	8,747
Panel B: Full-time			
RD estimate	−0.003	−0.041	−0.001
	(0.025)	(0.035)	(0.047)
Control mean	0.053	0.240	0.545
Observations	3,740	3,740	3,740
Panel C: Part-time			
RD estimate	0.012	−0.066**	0.004
	(0.028)	(0.027)	(0.041)
Control mean	0.074	0.192	0.565
Observations	5,007	5,007	5,007

Notes: The table reports RD estimates of equation (1), pooled over 2015–2018. Antidepressants and Pain medication are indicators for any reimbursed prescription in the relevant therapeutic class during the calendar year (ATC codes N06A, and N02A, N02B, or M01A, respectively). Healthcare expenditure (>p50) is an indicator for total healthcare expenditure above the median of the 2013 distribution in the analysis sample. Outcomes are observed for all teachers in the 2013 sample and are therefore not conditional on remaining in teaching. Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below. The bandwidth is 240 days around the August 1, 1962, cutoff, with a local linear specification and a uniform kernel. Standard errors, in parentheses, are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 7: Long-run effect of eligibility on retention and destination of exits, by contract type, 2015–2024

	Retention	Non-education	Support staff	Total employment
	(1)	(2)	(3)	(4)
Panel A: All				
RD estimate	0.005	0.004	0.008	0.012
	(0.024)	(0.010)	(0.012)	(0.017)
Control mean	0.83	0.030	0.024	0.890
Observations	21,689	21,689	21,689	21,689
Panel B: Full-time				
RD estimate	0.063*	−0.007	−0.016	0.036*
	(0.033)	(0.013)	(0.021)	(0.020)
Control mean	0.834	0.028	0.040	0.910
Observations	9,262	9,262	9,262	9,262
Panel C: Part-time				
RD estimate	−0.037	0.011	0.028**	−0.002
	(0.034)	(0.014)	(0.014)	(0.026)
Control mean	0.826	0.031	0.011	0.871
Observations	12,427	12,427	12,427	12,427
Panel D: Difference (full-time − part-time)				
RD estimate	0.100**	−0.017	−0.044*	0.038
	(0.047)	(0.019)	(0.025)	(0.033)
Observations	21,689	21,689	21,689	21,689

Notes: The table reports RD estimates of equation (1), pooled over 2015–2024, so that the window includes the years from 2019 onward in which the control cohort is also eligible. Outcomes are defined as in table 3. Panel D reports the difference between the full-time and part-time estimates. Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below. The bandwidth is 240 days around the August 1, 1962, cutoff, with a local linear specification and a uniform kernel. Standard errors, in parentheses, are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 8: Long-run effect of eligibility on total teaching supplied, by contract type, 2016–2024

	Unconditional actual working hours (1)
Panel A: All	
RD estimate	−0.021
	(0.024)
Control mean	0.670
Observations	19,446
Panel B: Full-time	
RD estimate	0.056
	(0.034)
Control mean	0.758
Observations	8,295
Panel C: Part-time	
RD estimate	−0.064**
	(0.029)
Control mean	0.590
Observations	11,151

Notes: The table reports RD estimates of equation (1), pooled over 2016–2024, so that the window includes the years from 2019 onward in which the control cohort is also eligible. The outcome is total teaching supplied, defined as in table 4. Full-time is defined as $\geq$ 0.95 FTE in 2013; part-time, below. The bandwidth is 240 days around the August 1, 1962, cutoff, with a local linear specification and a uniform kernel. Standard errors, in parentheses, are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 9: Long-run effect of eligibility on health outcomes, by contract type, 2015–2024

	Antidepressants	Pain medication	Healthcare expenditure (>p50)
	(1)	(2)	(3)
Panel A: All			
RD estimate	−0.001	−0.034*	0.023
	(0.019)	(0.017)	(0.028)
Control mean	0.073	0.184	0.581
Observations	21,689	21,689	21,689
Panel B: Full-time			
RD estimate	−0.020	−0.027	0.003
	(0.025)	(0.027)	(0.042)
Control mean	0.070	0.197	0.576
Observations	9,262	9,262	9,262
Panel C: Part-time			
RD estimate	0.012	−0.037*	0.038
	(0.027)	(0.022)	(0.037)
Control mean	0.075	0.172	0.580
Observations	12,427	12,427	12,427

Notes: The table reports RD estimates of equation (1), pooled over 2015–2024, so that the window includes the years from 2019 onward in which the control cohort is also eligible. Outcomes are defined as in table 6 and are not conditional on remaining in teaching. Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below. The bandwidth is 240 days around the August 1, 1962, cutoff, with a local linear specification and a uniform kernel. Standard errors, in parentheses, are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 10: Effect of eligibility on BAPO take-up and retention: lowest-scale teachers and support staff

	Teachers, scale LB		Support staff	
	BAPO take-up	Retention	BAPO take-up	Retention
	(1)	(2)	(3)	(4)
Panel A: All				
RD estimate	0.249***	0.004	0.272***	0.010
	(0.041)	(0.033)	(0.036)	(0.018)
Control mean	0.055	0.876	0.029	0.968
Observations	4,732	4,732	5,261	5,261
Panel B: Full-time				
RD estimate	0.393***	0.154***	0.331***	0.007
	(0.080)	(0.052)	(0.057)	(0.022)
Control mean	0.084	0.827	0.049	0.971
Observations	1,636	1,636	2,327	2,327
Panel C: Part-time				
RD estimate	0.204***	−0.064	0.231***	0.015
	(0.046)	(0.041)	(0.045)	(0.027)
Control mean	0.038	0.902	0.013	0.966
Observations	3,096	3,096	2,934	2,934

Notes: The table reports RD estimates of equation (1) for two samples: teachers in salary scale LB in 2013 (cols. 1–2) and administrative and educational-support staff employed in secondary education in 2013, excluding teaching staff (cols. 3–4). BAPO take-up is an indicator for participating in the program at any point during the year. Retention is an indicator for appearing in the personnel register in October in the same staff category, as a teacher in column 2 and as support staff in column 4. Both outcomes are pooled over 2015–2018. Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below. The bandwidth is 300 days around the August 1, 1962, cutoff, wider than the 240 days of the main tables to gain precision in these smaller samples; full-time estimates at 240 days are similar. All specifications are local linear with a uniform kernel. Standard errors, in parentheses, are clustered at the individual level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 11: Cost of covering a vacancy-year, by coverage mode

Coverage mode	Weight	Money (salaries)	Quality loss (SD)	Midpoint
Open vacancy, patched out-of-field	0.31	−0.15	0.05–0.11	0.08
Qualified colleague, extra hours	0.12	−0.15	$\frac{1}{3}\times$(0.02–0.06)	0.013
Out-of-field or unqualified colleague	0.39	−0.15	0.05–0.11	0.08
Agency or self-employed, qualified	0.11	+0.95	0	0
Agency or self-employed, out-of-field	0.07	+0.95	0.05–0.11	0.08
Weighted average		+0.05		0.06

Notes: Money is the cost of cover minus the senior's salary, in units of €51,300: a seconded teacher costs €113 per hour against €58 for a directly employed teacher in the same scale and step (Algemene Rekenkamer, 2024); colleagues covering internally are assumed to earn 0.85 of senior pay. Weights are shares of the total teacher shortage in October 2024 (Ministerie van Onderwijs, Cultuur en Wetenschap, 2024, Table 11); the remaining 0.01 (merged classes) is omitted. Open vacancies are assigned to out-of-field cover, since their lessons are patched internally while they remain unfilled. Quality loss is in standard deviations of student test scores per year of exposure, relative to an experienced teacher certified in the subject. Out-of-field: students of a teacher not certified in the subject score 0.07 SD lower on average (0.05 if certified in a related subject) and 0.10–0.11 lower in mathematics and English (Clotfelter et al., 2010, Tables 4–5). Early career: first-year teachers are 0.05–0.13 SD less effective than experienced teachers and the gap closes over the following four years (Rivkin et al., 2005, Table VII), (Clotfelter et al., 2010, Table 4), (Papay and Kraft, 2015, Table 2), so a teacher in her first five years is on average 0.02–0.06 SD less effective; one third of covering colleagues are assumed to be in their first five years. Experienced colleagues are assumed as effective as the senior, following Rivkin et al. (2005) and Clotfelter et al. (2010); Papay and Kraft (2015) find gains of 0.03–0.05 SD in mathematics between years 5 and 15. Weighted averages use midpoints.

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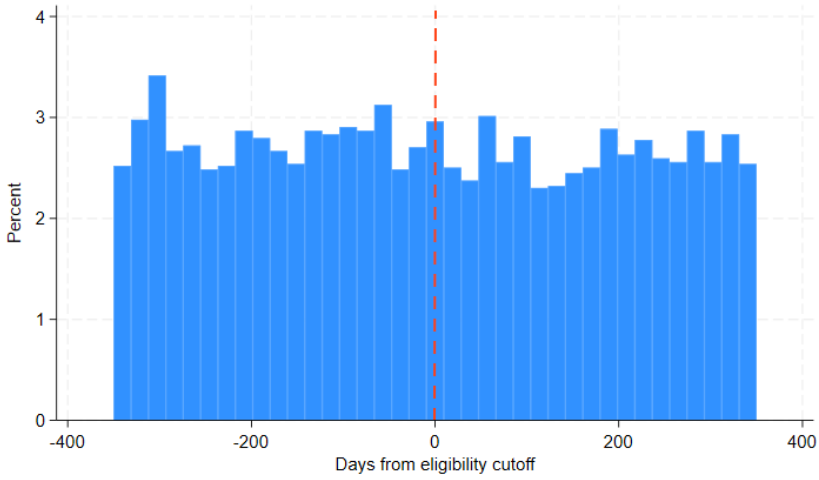
A Additional Tables and Figures

Table A.1: Effect of eligibility on attrition from the population register

	All	Full-time	Part-time
	(1)	(2)	(3)
RD estimate	0.010	0.009	0.011
	(0.010)	(0.012)	(0.014)
Control mean	0.009	0.007	0.011
Observations	9,076	3,852	5,224

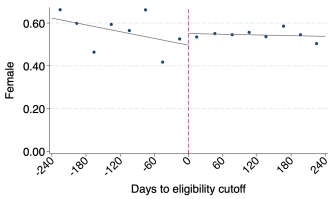
Notes: The table reports RD estimates of equation (1), pooled over 2015–2018. The outcome is an indicator equal to one if the teacher ceases to be registered as a Dutch resident in the year. The sample consists of the 2,278 teachers of the main sample before the residency restriction described in Appendix B; teachers are not observed after they cease to be resident. Full-time is defined as $\geq$ 0.95 FTE in 2013; part-time, below. The bandwidth is 240 days around the August 1, 1962, cutoff, with a local linear specification and a uniform kernel. Standard errors, in parentheses, are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Figure A.1: Manipulation test of the running variable

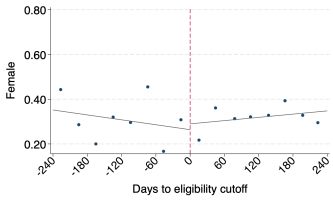


Notes: The figure plots the density of the running variable (days from the 1 August 1962 cutoff) for the analysis sample. The vertical dashed line marks the cutoff.

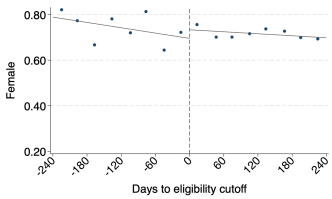
Figure A.2: Covariate balance: demographic characteristics



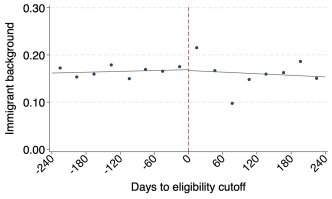
(a) Female, full sample



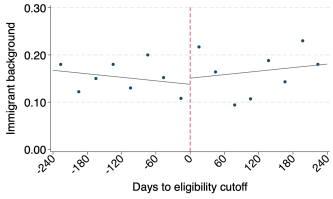
(b) Female, full-time



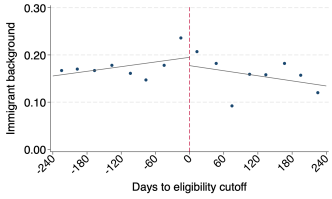
(c) Female, part-time



(d) Immigrant background, full sample



(e) Immigrant background, full-time



(f) Immigrant background, part-time

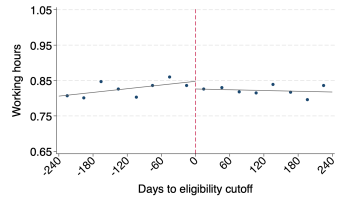
Notes: Each panel plots the predetermined covariate indicated in the panel title against date of birth relative to the 1 August 1962 cutoff, for the full sample and separately for full- and part-time teachers, with evenly spaced 30-day bins and separate linear fits on each side; the bandwidth is 240 days with a uniform kernel. Full-time is defined as ≥ 0.95 FTE in 2013; part-time below.

Table A.2: Covariate balance at the cutoff: predetermined characteristics and pre-reform outcomes, by contract type

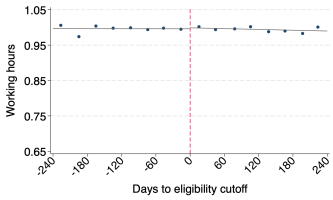
	All		Full-time		Part-time	
	RD est.	Std. err.	RD est.	Std. err.	RD est.	Std. err.
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Demographics and labor market						
Female	0.062	(0.042)	0.032	(0.058)	0.046	(0.051)
Immigrant background	−0.013	(0.032)	−0.013	(0.046)	−0.015	(0.044)
Contractual working hours	−0.021	(0.016)	0.000	(0.006)	−0.012	(0.018)
Monthly salary (euros)	−82.6	(109.1)	42.9	(92.7)	−16.1	(116.4)
Panel B: Health						
Antidepressants	0.012	(0.020)	−0.013	(0.023)	0.029	(0.030)
Pain medication	−0.015	(0.026)	−0.011	(0.040)	−0.019	(0.034)
Healthcare expenditure (>p50)	−0.024	(0.036)	−0.002	(0.053)	−0.049	(0.048)
Observations	4,389		1,877		2,512	

Notes: The table reports RD estimates of equation (1) at the August 1, 1962, cutoff using pre-reform data, pooled over 2012–2013. Female and Immigrant background are time invariant and are estimated on the same person-years as the other rows for comparability. Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below. The bandwidth is 240 days, with a local linear specification and a uniform kernel. Standard errors, in parentheses, are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

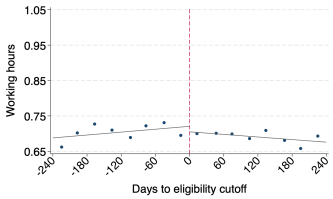
Figure A.3: Covariate balance: pre-reform employment and health outcomes



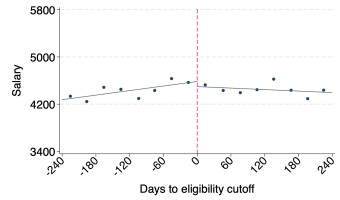
(a) Contractual working hours, full sample



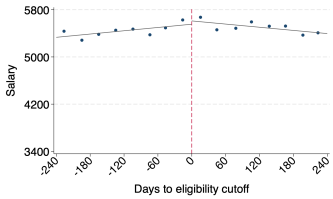
(b) Contractual working hours, full-time



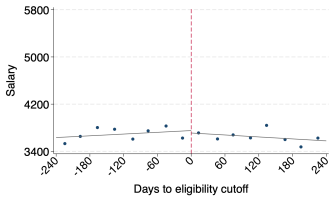
(c) Contractual working hours, part-time



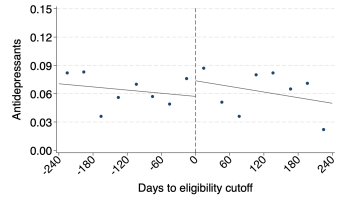
(d) Monthly salary, full sample



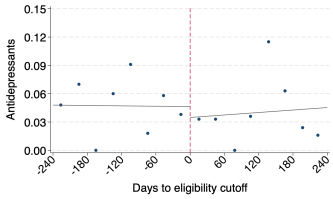
(e) Monthly salary, full-time



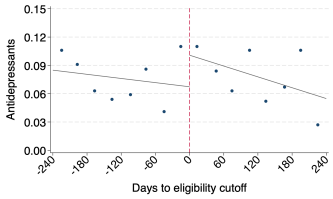
(f) Monthly salary, part-time



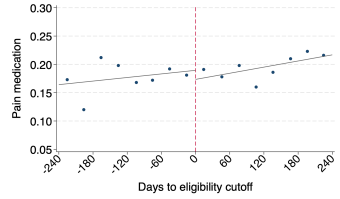
(g) Antidepressants, full sample



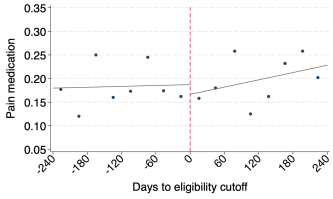
(h) Antidepressants, full-time



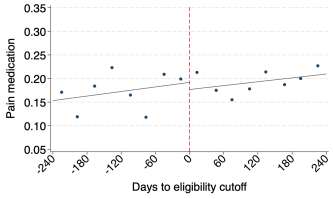
(i) Antidepressants, part-time



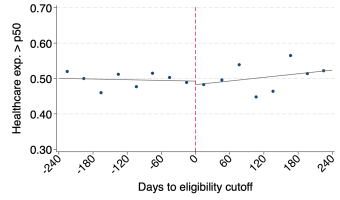
(j) Pain medication, full sample



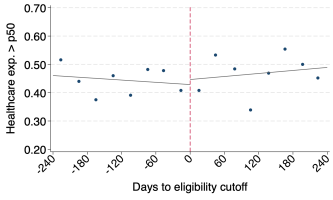
(k) Pain medication, full-time



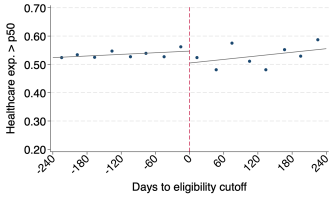
(l) Pain medication, part-time



(m) Healthcare expenditure (>p50), full sample



(n) Healthcare expenditure (>p50), full-time



(o) Healthcare expenditure (>p50), part-time

Notes: Each panel plots the pre-reform value of the variable indicated in the panel title against date of birth relative to the 1 August 1962 cutoff, for the full sample and separately for full- and part-time teachers, with evenly spaced 30-day bins and separate linear fits on each side; the bandwidth is 240 days with a uniform kernel. Full-time is defined as ≥ 0.95 FTE in 2013; part-time below.

Table A.3: Local randomization estimates of the effect of eligibility on BAPO take-up and retention

	Window (days from cutoff)		
	±30	±60	±90
Panel A: All teachers			
BAPO take-up	0.423***	0.409***	0.396***
	(0.000)	(0.000)	(0.000)
Retention	0.040	0.016	0.019
	(0.205)	(0.433)	(0.267)
Obs. (left/right)	137 / 144	276 / 282	406 / 436
Panel B: Full-time teachers			
BAPO take-up	0.482***	0.423***	0.460***
	(0.000)	(0.000)	(0.000)
Retention	0.061	0.041	0.055**
	(0.177)	(0.176)	(0.016)
Obs. (left/right)	65 / 60	131 / 121	186 / 185
Panel C: Part-time teachers			
BAPO take-up	0.385***	0.400***	0.350***
	(0.000)	(0.000)	(0.000)
Retention	0.027	−0.002	−0.008
	(0.574)	(0.978)	(0.743)
Obs. (left/right)	72 / 84	145 / 161	220 / 251

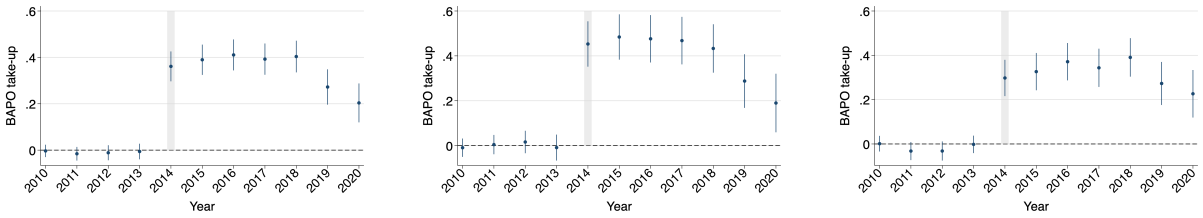
Notes: The table reports local randomization estimates of the effect of eligibility within symmetric windows of ±30, ±60, and ±90 days around the August 1, 1962, cutoff, following Cattaneo et al. (2016). Each cell reports the difference in mean outcomes between teachers born before and on or after the cutoff, pooled over 2015–2018; finite-sample randomization p-values, based on 10,000 permutations of eligibility within the window, are in parentheses. Obs. (left/right) is the number of teachers on each side of the cutoff within the window. Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A.4: Effect of eligibility on permanent retention in teaching

	All	Full-time	Part-time
	(1)	(2)	(3)
RD estimate	−0.002	0.060**	−0.049*
	(0.018)	(0.024)	(0.025)
Control mean	0.960	0.936	0.978
Observations	8,747	3,740	5,007

Notes: The table reports RD estimates of equation (1), pooled over 2015–2018. The outcome is an absorbing measure of retention that treats a teacher as having exited in year t only if she never reappears in the teacher register thereafter. Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below. The bandwidth is 240 days around the August 1, 1962, cutoff, with a local linear specification and a uniform kernel. Standard errors, in parentheses, are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Figure A.4: Year-by-year effect of eligibility on BAPO take-up, 2010–2020



(a) All teachers (b) Full-time (c) Part-time

Notes: Each panel plots year-by-year RD estimates of equation (1) with 95 percent confidence intervals based on standard errors clustered at the teacher level. The shaded band marks 2014, excluded because the treatment group is only partially exposed in that year. Full-time is defined as ≥ 0.95 FTE in 2013; part-time below.

Table A.5: Effect of eligibility on retention, by 2013 contract size

Panel A: Contract-size bins			
	≤ 0.70 FTE	0.71–0.94 FTE	≥ 0.95 FTE
	(1)	(2)	(3)
RD estimate	−0.068	0.014	0.061**
	(0.050)	(0.031)	(0.028)
Control mean	0.918	0.916	0.900
Observations	2,210	2,797	3,740
Panel B: Cumulative full-time thresholds			
	≥ 0.70 FTE	≥ 0.85 FTE	≥ 0.95 FTE
	(1)	(2)	(3)
RD estimate	0.045**	0.060**	0.061**
	(0.021)	(0.025)	(0.028)
Control mean	0.907	0.906	0.900
Observations	6,731	4,392	3,740

Notes: Each cell reports an RD estimate of equation (1) for retention in teaching, pooled over 2015–2018, on the subsample defined by 2013 contract size in the column heading. Panel A uses mutually exclusive bins; panel B uses cumulative thresholds, so column 3 of each panel reproduces the full-time estimate of table 3. The bandwidth is 240 days around the August 1, 1962, cutoff, with a local linear specification and a uniform kernel. Standard errors, in parentheses, are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A.6: Effect of eligibility, by contract type and gender

	Women			Men		
	All	Full-time	Part-time	All	Full-time	Part-time
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Extensive margin						
BAPO take-up	0.386***	0.458***	0.367***	0.408***	0.476***	0.282***
	(0.035)	(0.074)	(0.039)	(0.042)	(0.052)	(0.069)
Retention	−0.012	0.052	−0.031	0.026	0.053*	−0.021
	(0.025)	(0.043)	(0.030)	(0.025)	(0.031)	(0.043)
Observations	5,978	1,492	4,486	5,036	3,294	1,742
Panel B: Working hours						
Actual working hours	−0.045	0.058	−0.074**	−0.016	0.013	−0.027
	(0.028)	(0.050)	(0.029)	(0.030)	(0.035)	(0.049)
Observations	4,463	1,114	3,349	3,755	2,456	1,299
Panel C: Health						
Antidepressants	0.005	0.012	0.003	0.006	0.003	0.012
	(0.026)	(0.046)	(0.031)	(0.021)	(0.025)	(0.039)
Pain medication	−0.045*	−0.122*	−0.018	−0.036	−0.015	−0.069
	(0.027)	(0.064)	(0.029)	(0.027)	(0.035)	(0.044)
Healthcare expenditure (>p50)	−0.004	−0.130*	0.034	0.000	0.033	−0.058
	(0.037)	(0.073)	(0.042)	(0.042)	(0.051)	(0.076)
Observations	5,978	1,492	4,486	5,036	3,294	1,742

Notes: The table reports RD estimates of equation (1), separately by contract type and gender, pooled over 2015–2018 (2016–2018 for actual working hours in panel B). Each cell corresponds to a separate regression. Outcomes are defined as in tables 2–6; actual working hours in panel B are unconditional on remaining in teaching, hence the smaller number of observations. Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below. The bandwidth is widened to 300 days around the August 1, 1962, cutoff to improve precision in the gender-specific subsamples, with a local linear specification and a uniform kernel. Standard errors, in parentheses, are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A.7: Effect of eligibility on additional health outcomes, by contract type

	Mental healthcare (broad) (1)	Mental healthcare expenditure (2)	GP expenditure (>p75) (3)
Panel A: All			
RD estimate	0.013	0.003	0.052**
	(0.020)	(0.007)	(0.026)
Control mean	0.083	0.025	0.276
Observations	8,747	8,747	8,747
Panel B: Full-time			
RD estimate	−0.005	−0.005	0.029
	(0.027)	(0.009)	(0.038)
Control mean	0.063	0.019	0.241
Observations	3,740	3,740	3,740
Panel C: Part-time			
RD estimate	0.024	0.008	0.065*
	(0.029)	(0.011)	(0.036)
Control mean	0.100	0.031	0.305
Observations	5,007	5,007	5,007

Notes: The table reports RD estimates of equation (1), pooled over 2015–2018. Mental healthcare (broad) is an indicator equal to one if the teacher fills an antidepressant prescription or has any reimbursed expenditure on generalist basic mental healthcare in the year. Mental healthcare expenditure is an indicator for any reimbursed generalist basic mental healthcare expenditure. GP expenditure (>p75) is an indicator for general-practitioner expenditure above the 75th percentile within the analysis sample. Outcomes are not conditional on remaining in teaching. Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below. The bandwidth is 240 days around the August 1, 1962, cutoff, with a local linear specification and a uniform kernel. Standard errors, in parentheses, are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A.8: Robustness: balanced panel

	RD estimate		
	All	Full-time	Part-time
	(1)	(2)	(3)
Panel A: Extensive margin			
BAPO take-up	0.394***	0.471***	0.342***
	(0.030)	(0.048)	(0.038)
Retention	0.004	0.060**	−0.036
	(0.020)	(0.028)	(0.028)
Observations	8,940	3,808	5,132
Panel B: Intensive margin			
Actual working hours	−0.039	0.039	-0.077***
	(0.024)	(0.032)	(0.028)
Observations	6,687	2,846	3,841
Panel C: Health			
Antidepressants	0.008	−0.005	0.016
	(0.019)	(0.025)	(0.027)
Pain medication	−0.054**	−0.041	−0.060**
	(0.021)	(0.035)	(0.027)
Healthcare expenditure (>p50)	0.000	−0.010	0.004
	(0.031)	(0.047)	(0.04)
Observations	8,940	3,808	5,132

Notes: The table reports RD estimates of equation (1) on the balanced panel, in which the 193 person-years observed in the employer–employee register under an education-sector employer but absent from the teacher register are imputed as teaching employment (app. B). Outcomes are defined as in tables 2–6 and pooled over 2015–2018 (2016–2018 for actual working hours in panel B). Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below. The bandwidth is 240 days around the August 1, 1962, cutoff, with a local linear specification and a uniform kernel. Standard errors, in parentheses, are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A.9: Robustness: baseline covariates included

	RD estimate		
	All	Full-time	Part-time
	(1)	(2)	(3)
Panel A: Extensive margin			
BAPO take-up	0.406***	0.468***	0.358***
	(0.030)	(0.049)	(0.038)
Retention	0.015	0.059**	−0.017
	(0.020)	(0.028)	(0.027)
Observations	8,747	3,740	5,007
Panel B: Intensive margin			
Actual working hours	−0.004	0.040	−0.039
	(0.020)	(0.032)	(0.024)
Observations	6,524	2,787	3,737
Panel C: Health			
Antidepressants	0.020	0.019	0.021
	(0.017)	(0.023)	(0.023)
Pain medication	−0.052**	−0.034	−0.066**
	(0.021)	(0.033)	(0.026)
Healthcare expenditure (>p50)	0.006	0.011	0.004
	(0.030)	(0.046)	(0.041)
Observations	8,747	3,740	5,007

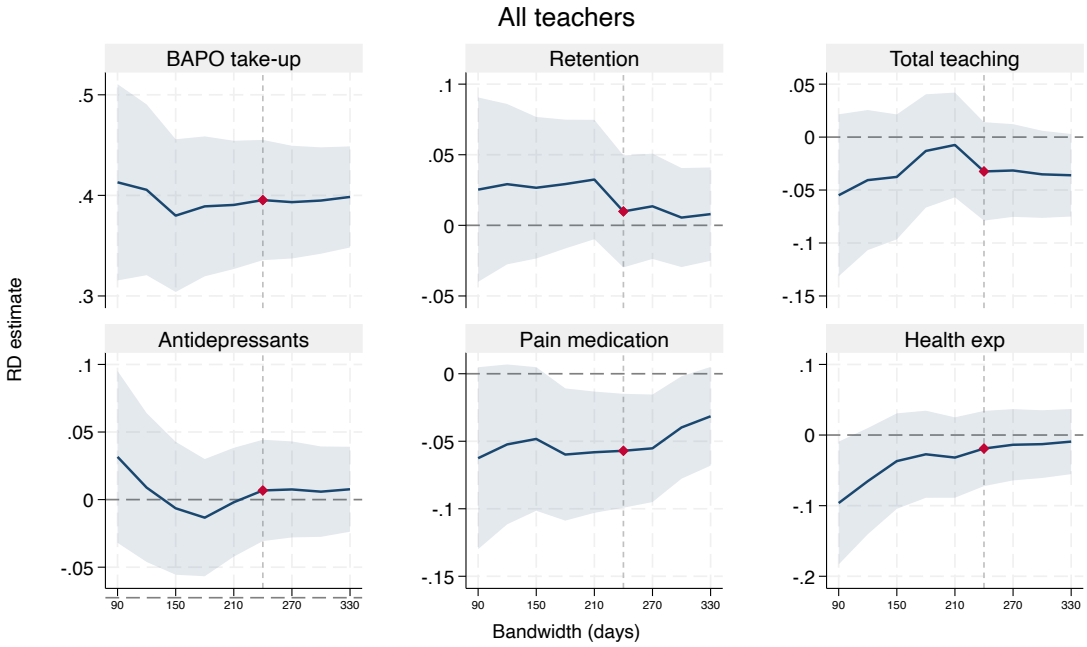
Notes: The table reports RD estimates of equation (1) including the full vector of baseline covariates: gender, immigrant background, contractual FTE, and antidepressant and pain-medication use, all measured in 2013. Outcomes are defined as in tables 2–6 and pooled over 2015–2018 (2016–2018 for actual working hours in panel B). Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below. The bandwidth is 240 days around the August 1, 1962, cutoff, with a local linear specification and a uniform kernel. Standard errors, in parentheses, are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A.10: Robustness: second-order polynomial in the running variable

	RD estimate		
	All	Full-time	Part-time
	(1)	(2)	(3)
Panel A: Extensive margin			
BAPO take-up	0.390***	0.406***	0.386***
	(0.046)	(0.074)	(0.058)
Retention	0.052*	0.083*	0.030
	(0.031)	(0.043)	(0.043)
Observations	8,747	3,740	5,007
Panel B: Intensive margin			
Actual working hours	−0.012	0.059	−0.033
	(0.036)	(0.047)	(0.044)
Observations	6,524	2,787	3,737
Panel C: Health			
Antidepressants	−0.003	−0.005	−0.006
	(0.030)	(0.041)	(0.044)
Pain medication	−0.064**	−0.032	−0.081**
	(0.032)	(0.053)	(0.039)
Healthcare expenditure (>p50)	−0.047	−0.048	−0.050
	(0.047)	(0.070)	(0.062)
Observations	8,747	3,740	5,007

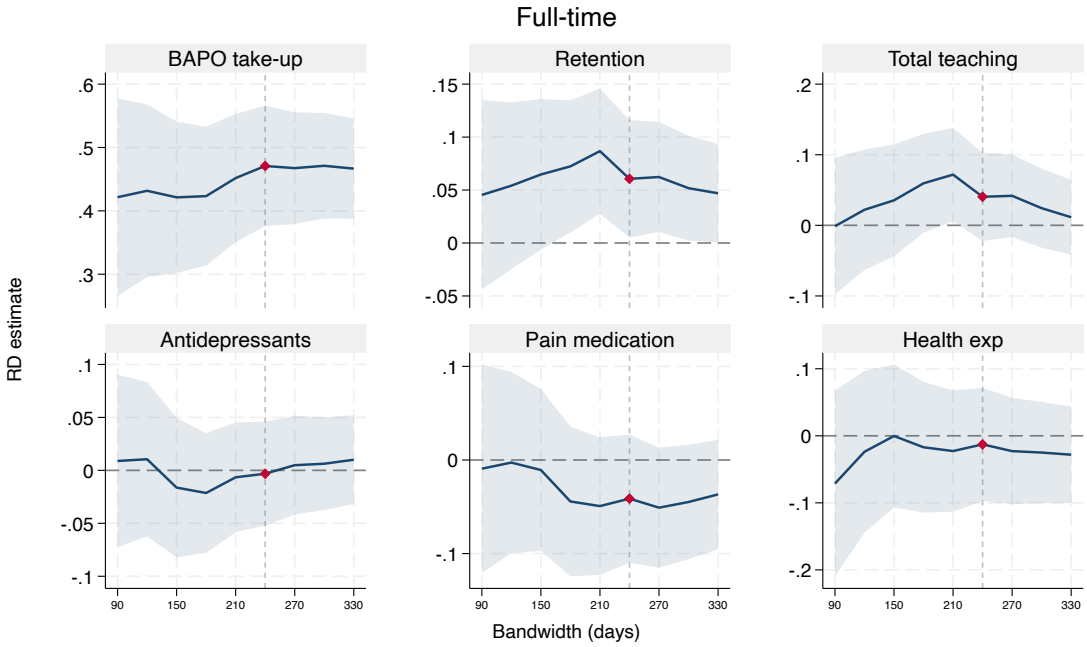
Notes: The table reports RD estimates of equation (1) allowing a second-order polynomial in the running variable on each side of the cutoff. Outcomes are defined as in tables 2–6 and pooled over 2015–2018 (2016–2018 for actual working hours in panel B); control means are reported in the main tables. Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below. The bandwidth is 240 days around the August 1, 1962, cutoff with a uniform kernel. Standard errors, in parentheses, are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Figure A.5: Bandwidth sensitivity, all teachers



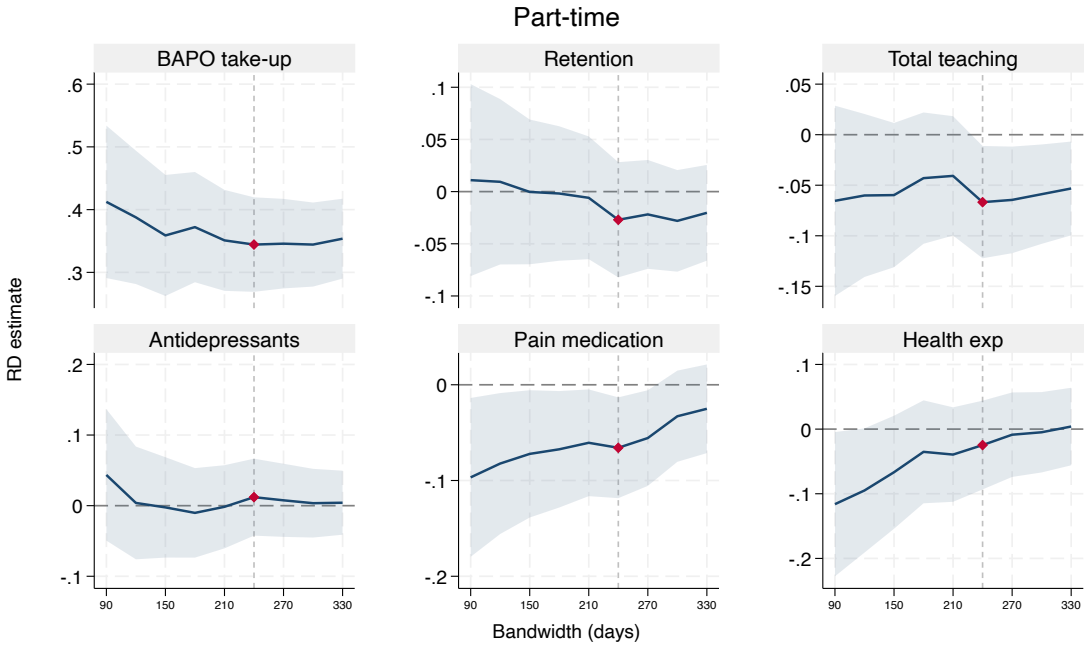
Notes: The figure plots RD estimates and 95 percent confidence intervals for the main outcomes across alternative bandwidths ranging from 90 to 330 days. Estimates correspond to equation (1) pooled over the post-reform period 2015–2018 (2016–2018 for the intensive-margin outcomes).

Figure A.6: Bandwidth sensitivity, full-time teachers



Notes: The figure plots RD estimates and 95 percent confidence intervals for the main outcomes across alternative bandwidths ranging from 90 to 330 days, for full-time teachers ($\geq$ 0.95 FTE in 2013). Estimates correspond to equation (1) pooled over the post-reform period 2015–2018 (2016–2018 for the intensive-margin outcomes).

Figure A.7: Bandwidth sensitivity, part-time teachers



Notes: The figure plots RD estimates and 95 percent confidence intervals for the main outcomes across alternative bandwidths ranging from 90 to 330 days, for part-time teachers (below 0.95 FTE in 2013). Estimates correspond to equation (1) pooled over the post-reform period 2015–2018 (2016–2018 for the intensive-margin outcomes).

Table A.11: Robustness: donut regression discontinuity estimates

	RD estimate		
	All	Full-time	Part-time
	(1)	(2)	(3)
Panel A: Extensive margin			
BAPO take-up	0.373***	0.463***	0.309***
	(0.040)	(0.063)	(0.050)
Retention	−0.009	0.058	−0.059*
	(0.025)	(0.036)	(0.035)
Observations	7,632	3,241	4,391
Panel B: Intensive margin			
Actual working hours	−0.027	0.047	−0.068*
	(0.030)	(0.042)	(0.035)
Observations	5,689	2,413	3,276
Panel C: Health			
Antidepressants	−0.004	−0.009	−0.000
	(0.024)	(0.033)	(0.034)
Pain medication	−0.034	−0.012	−0.050
	(0.029)	(0.047)	(0.036)
Healthcare expenditure (>p50)	0.039	0.041	0.035
	(0.041)	(0.064)	(0.054)
Observations	7,632	3,241	4,391

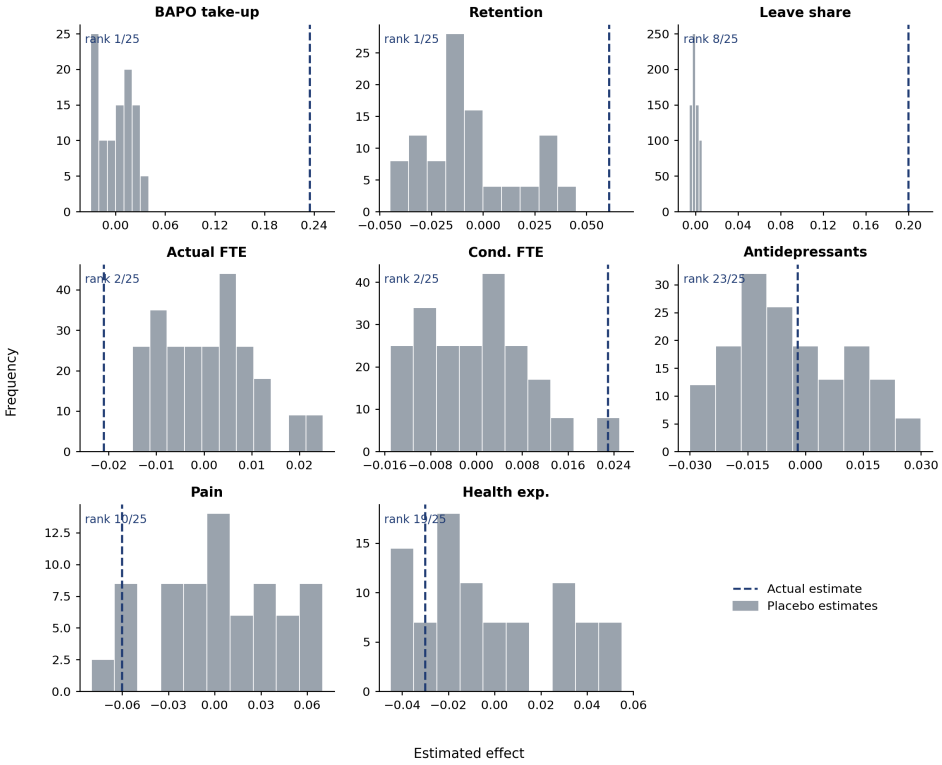
Notes: The table reports RD estimates of equation (1) excluding teachers born within one month of the August 1, 1962, cutoff. Outcomes are defined as in tables 2–6 and pooled over 2015–2018 (2016–2018 for actual working hours in panel B). Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below. The bandwidth is 240 days, with a local linear specification and a uniform kernel. Standard errors, in parentheses, are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A.12: Robustness: MSE-optimal bandwidths with robust bias-corrected inference

	RD estimate (1)	Robust std. err. (2)	Robust p-value (3)	h_ℓ (4)	h_r (5)	N_ℓ (6)	N_r (7)
Panel A: All							
BAPO take-up	0.391***	(0.039)	<0.001	253.5	253.5	4,562	4,677
Retention	0.023	(0.023)	0.213	306.5	306.5	5,463	5,787
Actual working hours	−0.024	(0.026)	0.417	345.6	345.6	4,583	4,907
Antidepressants	0.006	(0.022)	0.671	327.8	327.8	5,848	6,247
Pain medication	−0.058**	(0.029)	0.019	209.4	209.4	3,731	3,852
Healthcare expenditure (>p50)	−0.006	(0.037)	0.498	309.3	309.3	5,503	5,838
Panel B: Full-time							
BAPO take-up	0.449***	(0.060)	<0.001	261.4	261.4	1,971	2,153
Retention	0.072**	(0.035)	0.017	260.6	260.6	1,967	2,137
Actual working hours	0.051*	(0.039)	0.090	244.9	244.9	1,385	1,462
Antidepressants	0.009	(0.026)	0.706	395.8	395.8	2,905	3,439
Pain medication	−0.035	(0.038)	0.290	355.2	355.2	2,656	3,052
Healthcare expenditure (>p50)	−0.017	(0.054)	0.756	306.8	306.8	2,271	2,615
Panel C: Part-time							
BAPO take-up	0.352***	(0.046)	<0.001	288.4	288.4	2,991	2,930
Retention	−0.013	(0.027)	0.672	437.9	437.9	4,414	4,493
Actual working hours	−0.046	(0.031)	0.124	359.0	359.0	2,738	2,776
Antidepressants	0.002	(0.031)	0.825	333.6	333.6	3,464	3,455
Pain medication	−0.088**	(0.039)	0.011	176.5	176.5	1,776	1,840
Healthcare expenditure (>p50)	0.002	(0.047)	0.992	319.4	319.4	3,355	3,296

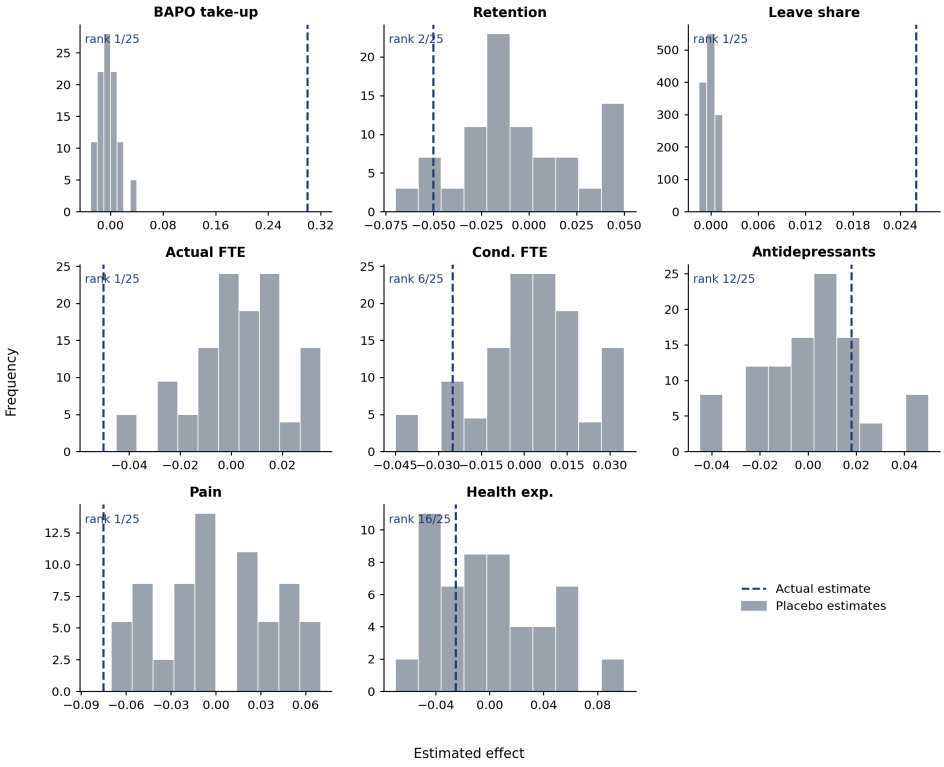
Notes: The table reports RD estimates at the MSE-optimal bandwidth selected separately by outcome and sample following Calonico et al. (2014), pooled over 2015–2018 (2016–2018 for actual working hours). Column 1 reports the conventional local linear estimate; columns 2 and 3 the robust bias-corrected standard error and p-value, on which significance levels are based. Columns 4 and 5 report the bandwidths in days to the left and right of the cutoff, which coincide under the common-bandwidth procedure; columns 6 and 7 the corresponding numbers of observations. Estimates use a triangular kernel, with standard errors clustered at the teacher level. Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Figure A.8: Placebo cutoffs, full-time teachers



Notes: Distribution of estimates of equation (1) at 25 placebo cutoffs 9 to 33 months after 1 August 1962, full-time teachers. Dashed line: estimate at the true cutoff. Rank: position of the true estimate among the placebo estimates, from smallest to largest.

Figure A.9: Placebo cutoffs, part-time teachers



Notes: Distribution of estimates of equation (1) at 25 placebo cutoffs 9 to 33 months after 1 August 1962, part-time teachers. Dashed line: estimate at the true cutoff. Rank: position of the true estimate among the placebo estimates, from smallest to largest.

Table A.13: Placebo-cutoff estimates and randomization-inference p-values

Outcome	True estimate	Placebo mean	Rank	p
Panel A: All teachers				
BAPO take-up	0.395	0.003	1	0.038
Retention	0.010	0.001	9	0.346
Total teaching supplied	−0.032	0.003	2	0.077
Antidepressants	0.007	0.000	9	0.346
Pain medication	−0.057	0.003	1	0.038
Healthcare expenditure (>p50)	0.003	0.006	13	0.500
Panel B: Full-time teachers				
BAPO take-up	0.471	0.004	1	0.038
Retention	0.061	−0.002	1	0.038
Total teaching supplied	0.040	0.002	6	0.231
Antidepressants	−0.003	0.000	13	0.500
Pain medication	−0.041	0.005	5	0.192
Healthcare expenditure (>p50)	−0.001	0.004	12	0.462
Panel C: Part-time teachers				
BAPO take-up	0.344	0.001	1	0.038
Retention	−0.027	0.003	3	0.115
Total teaching supplied	−0.067	0.003	1	0.038
Antidepressants	0.012	0.001	8	0.308
Pain medication	−0.066	0.001	3	0.115
Healthcare expenditure (>p50)	0.004	0.007	14	0.538

Notes: The table re-estimates equation (1) at 25 placebo cutoffs, monthly birth-date thresholds from May 1963 to May 1965 at which eligibility does not change; all placebo cohorts are deferred to age 57, and the nine-month gap after the true cutoff ensures that no placebo window overlaps the true discontinuity. True estimate is the RD estimate at the August 1, 1962, cutoff and Placebo mean the mean of the 25 placebo estimates. Rank is the position of the true estimate among the 26 estimates, counted from the largest for positive estimates and from the smallest for negative ones; p is the corresponding one-sided randomization-inference p-value, rank/26 (Ganong and Jäger, 2018), whose smallest attainable value is $1/26 \approx 0.038$. Outcomes are pooled over 2015–2018 (2016–2018 for total teaching supplied) with a 240-day bandwidth, a local linear specification, and a uniform kernel. Health outcomes are not conditional on remaining in teaching. Full-time is defined as ≥ 0.95 FTE in 2013; part-time, below.

Table A.14: Placebo difference-in-differences: testing for a control-cohort response

	Retention	Antidepressants	Pain medication
	(1)	(2)	(3)
Panel A: All			
Cohort 1 × Post	−0.019*	0.002	−0.003
	(0.010)	(0.007)	(0.011)
Cohort 2 × Post	−0.008	0.002	−0.010
	(0.010)	(0.007)	(0.012)
Observations	25,019	25,019	25,019
Panel B: Full-time			
Cohort 1 × Post	−0.010	−0.001	0.025
	(0.015)	(0.008)	(0.017)
Cohort 2 × Post	0.002	0.000	−0.026
	(0.014)	(0.009)	(0.018)
Observations	10,799	10,799	10,799
Panel C: Part-time			
Cohort 1 × Post	−0.026*	0.004	−0.024
	(0.014)	(0.011)	(0.015)
Cohort 2 × Post	−0.016	0.004	0.003
	(0.014)	(0.010)	(0.016)
Observations	14,220	14,220	14,220

Notes: Panel A pools all teachers; Panel B restricts the sample to full-time teachers (≥ 0.95 FTE in 2013); Panel C to part-time teachers. Each column reports a difference-in-differences across three cohorts excluded from the transitional BAPO arrangement, estimated over 2011–2018. Cohort 1 is the RD control cohort (born August 1962–March 1963), which missed eligibility most narrowly; Cohort 2 is born one year later (August 1963–March 1964); the omitted reference cohort is born two years later (August 1964–March 1965). Post equals one for 2015–2018. All specifications include cohort and year fixed effects. If narrowly missing the cutoff generates a disappointment response, it should be strongest for Cohort 1 and attenuate with distance from the cutoff. Coefficients not statistically distinguishable from zero, and showing no gradient across cohorts, indicate no such response. All outcomes are defined as in Tables 3 and 6. Standard errors in parentheses are clustered at the teacher level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.15: Difference-in-differences estimates of school exposure on incumbent teachers' working hours, by age group

	Young (< 35) (1)	Middle-aged (35–49) (2)	Older (≥ 50) (3)
Exposure (10 pp) × Post	0.0175***	0.0017	0.0002
	(0.0047)	(0.0024)	(0.0024)
Observations	163,124	243,079	199,663
Teachers	22,707	32,379	30,778

Notes: Each column reports the coefficient on $\pi_s \times \text{Post}_t$ from a teacher fixed-effects regression of actual working FTE (set to zero for teachers no longer employed as teaching staff) on school exposure interacted with a post-2014 indicator, controlling for the 2013 FTE-weighted shares of the deferred, always-eligible, and younger cohorts and school size, each interacted with Post. Coefficients and standard errors are scaled to a 10 percentage point increase in the exposure share. Age is defined in 2013. Standard errors in parentheses are clustered at the teacher level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Sample Construction

My primary data source is the administrative registers of public-sector education employees in the Netherlands, provided by DUO. Spanning 2006 to 2024, these registers record every individual employment relationship as observed in October of each year. I begin from all individuals employed in secondary education in 2013 and born within my estimation bandwidth (December 1961 to March 1963), yielding 3,854 job records for 3,787 individuals. Jobs are classified as teaching (2,407 records, 62.5 percent), educational support and administration (1,307 records, 33.9 percent), management (138 records, 3.6 percent), or other (2 records, 0.1 percent).

The main analysis focuses on teachers, defined as all individuals holding at least one teaching job in 2013. This selection retains 2,412 job records for 2,362 individuals: the 2,407 teaching jobs plus five nonteaching jobs held by these same teachers. The 50 additional records (2.1 percent) belong to teachers holding multiple jobs, either within the same school or across schools. I aggregate them to a single record per teacher, summing FTE and salary across jobs and assigning each teacher to the school accounting for the largest number of hours, and I apply the same rule in all subsequent years. To validate the school-reported records, I link them to the employer–employee register (SPOLISBUS), drawn from firm tax records. Seven individuals appear in the teacher register but not in the tax data. I treat these as school reporting errors, most plausibly teachers no longer holding a formal contract with the school, and drop them, leaving 2,355 teachers. The registers also capture individuals whose teaching contract sits alongside larger employment (or self-employment) elsewhere. Because the analysis concerns teaching careers, I restrict the sample to teachers for whom teaching was the main source of income in 2013. This drops 77 individuals, leaving the 2,278 teachers of the main analysis.

I then construct a balanced panel over 2015–18 for these 2,278 teachers, for a total of 9,112 person-year observations. Over this period, 43 individuals cease to be Dutch residents, chiefly through death. Because the health data cover only residents insured under the mandatory basic health-insurance scheme, I drop these individuals entirely, including the years in which they were still resident, so that labor-market and health outcomes are estimated on an identical sample. This leaves 8,940 observations on 2,235 teachers. An RD estimate confirms that eligibility does not affect the probability of remaining resident (Table A.1), so the restriction is unlikely to introduce differential selection at the cutoff.

Finally, 193 of the remaining person-years (2.2 percent) appear in the employer–employee data under the education sector but not in the teacher register. This typically

reflects delayed school reporting to DUO, which the October snapshot misses: an individual may appear continuously in the employer–employee data under the same encrypted firm identifier while being absent from the teacher register in a single year between 2010 and 2018. Because retaining these person-years would require assuming continued employment as a teacher, the main specification drops them, yielding a final, unbalanced estimation panel of 8,747 observations. As a robustness check, I instead impute them as teaching employment, taking the job function from the teacher register and hours and salary from the employer–employee data; the results are unchanged (Table A.8).

C The Netherlands Working Conditions Survey (NEA)

The NEA (*Netherlands Working Conditions Survey*) is an annual survey of employees aged 15–74 working in the Netherlands, excluding the self-employed, conducted by TNO and Statistics Netherlands in cooperation with the Ministry of Social Affairs and Employment. It covers working conditions, job content, and employment terms. The survey is a repeated cross-section, and from 2017 onward the sample expands substantially, increasing the number of respondents in occupations such as teaching. Throughout, I restrict attention to secondary-school teachers aged 48–57 and born in or after August 1962, so that all face the post-reform regime without transitional BAPO access, and I pool all available waves (2014–2024). [34]

I use these data to compare job strain and the willingness to reduce working hours. The comparison between full-time and part-time teachers (Table C.1) uses all NEA respondents identified as secondary-school teachers in this age range; linking them to the analysis sample would leave too little overlap. The comparison between teachers and support staff (Table C.2) instead links NEA respondents through unique personal identifiers to the DUO personnel register, restricting to personnel employed in secondary education. Teaching and administrative and educational-support status is taken directly from the register, since the NEA itself cannot distinguish the two.

Variables are defined as follows. *Willing to work longer with fewer hours* equals one if the respondent reports that working fewer hours or fewer days per week is a circumstance under which they would be willing to continue working to a higher age. The item belongs to a multiple-response battery on conditions for working longer and is not fielded in 2014, 2016, 2018, or 2020, which accounts for its smaller sample size. *High work pressure* equals

[34] Figure 1 is the exception: it compares all secondary-school teachers with the Dutch workforce by age group over 50–65.

one if the respondent scores at least 2.5 on a three-item scale (working very fast, doing a great deal of work, working extra hard; 1 = never to 4 = always). *Physically demanding work* is a derived indicator based on exerting force, vibrations, awkward postures, and repetitive movements. *Emotionally demanding work* equals one if the respondent scores at least 2.5 on a three-item scale (emotionally difficult situations, emotionally demanding work, emotional involvement; same scale).

Table C.1: Workload and hours preferences of full- and part-time teachers (NEA)

	Part-time		Full-time		
	Mean	N	Mean	N	p-value
	(1)	(2)	(3)	(4)	(5)
Willing to work longer with fewer hours	0.657	1,459	0.738	690	0.000
High work pressure	0.484	2,209	0.520	981	0.061
Physically demanding work	0.101	2,198	0.132	980	0.012
Emotionally demanding work	0.263	1,887	0.282	794	0.304

Notes: The table reports means of NEA survey indicators for secondary-school teachers aged 48–57 and born in or after August 1962, pooled over 2014–2024. Variables are defined in the text. Column (5) reports the p-value of a two-sample test of equal means. Full-time status is defined from contractual weekly hours, approximating the ≥ 0.95 FTE definition used in the administrative analysis. The number of observations varies because of item non-response and because *willing to work longer with fewer hours* is not fielded in 2014, 2016, 2018, and 2020.

Table C.2: Working conditions and characteristics of lowest-scale teachers and support staff (NEA linked to the personnel register)

	Teachers, scale LB		Support staff		
	Mean	N	Mean	N	p-value
	(1)	(2)	(3)	(4)	(5)
Panel A: Working conditions (NEA)					
Willing to work longer with fewer hours	0.667	102	0.485	241	0.002
High work pressure	0.512	242	0.374	487	<0.001
Physically demanding work	0.138	240	0.243	489	<0.001
Emotionally demanding work	0.326	239	0.096	470	<0.001
Panel B: Characteristics in 2013					
Contractual working hours (FTE)	0.795	243	0.800	489	0.764
Monthly salary (euros)	2,882	243	2,403	489	<0.001
Female	0.584	243	0.603	489	0.624
Immigrant background	0.210	243	0.104	489	<0.001
Antidepressants	0.070	243	0.061	489	0.655
Pain medication	0.218	243	0.198	489	0.534

Notes: The table reports means of NEA survey indicators (Panel A) and of 2013 register and health-insurance characteristics (Panel B) for NEA respondents linked to the DUO personnel register, restricted to secondary-education personnel aged 48–57 and born in or after August 1962, pooled over 2014–2024. Columns (1)–(2) are restricted to teachers in salary scale LB; staff category is taken from the register. Column (5) reports the p-value of a two-sample test of equal means. Variables are defined as in Table C.1 and Table 1. The number of observations varies because of item non-response and because *willing to work longer with fewer hours* is not fielded in 2014, 2016, 2018, and 2020.

D Conditional Working hours

Table D.1 and Figure D.1 report effects on actual and contractual working hours, conditional on remaining a teacher. Eligibility reduces actual hours by 0.023 FTE among full-time teachers and by 0.048 FTE among part-time teachers. Because compliers in both groups use the same share of their entitlement, the smaller reduction in actual hours among full-time teachers must reflect a response in contractual hours. Consistent with this interpretation, contractual hours increase by 0.023 FTE at the cutoff among full-time

teachers.

This increase cannot reflect eligible teachers expanding their contracts: the collective agreement caps contracts at 1.0 FTE while BAPO leave is being taken (VO-raad, 2014). Nor can it be explained by changes in the composition of the surviving teacher pool. The retention response can account for at most 0.014 FTE of the increase, and only under the extreme assumption that every marginal stayer holds the largest permissible contract and takes no leave.[35] The remaining increase therefore reflects ineligible full-time teachers reducing their contracts at their own expense.

This pattern suggests that some full-time teachers are willing to reduce their workload even when doing so is costly, while the teachers deterred by that cost are precisely those whom eligibility induces to remain. The magnitudes are consistent with this interpretation. Thus, part of the subsidy appears to finance reductions in working time that some eligible teachers would otherwise have purchased themselves.

Part-time teachers show no comparable contractual response. Their contractual-hours estimate is negative and imprecisely estimated, while actual hours fall by 0.048 FTE. This pattern is consistent with funded leave passing through to actual working hours for part-time teachers, rather than subsidizing reductions in contractual hours that they would otherwise have financed themselves.

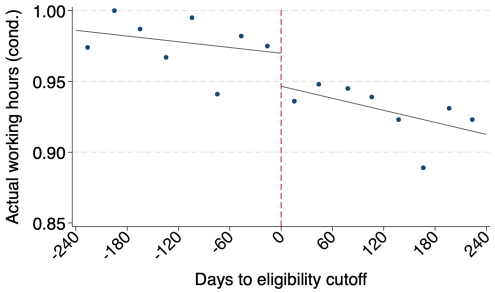
[35]Marginal stayers make up a share $s = \Delta/S_1 = 0.061/0.961 \approx 0.063$ of eligible full-time survivors, where $\Delta = 0.061$ is the retention effect and $S_1 = 0.900+0.061 = 0.961$ the implied retention rate of eligible teachers. Their maximum contribution to mean contractual hours is therefore $s\,(\bar{C}_{\text{marginal}} - \bar{C}_{\text{infra}}) \leq 0.063 \times (1.2 - 0.983) \approx 0.014$ FTE, where $\bar{C}_{\text{infra}} = 0.983$ is the control mean and 1.2 FTE is the maximum contract for a teacher not taking leave. The same share bounds any compositional effect on the leave-share estimates in Section 5 at $0.063 \times 0.102 \approx 0.006$, too small to affect the conclusion that takers exhaust the entitlement.

Table D.1: Net effect of eligibility on working hours, by employment group

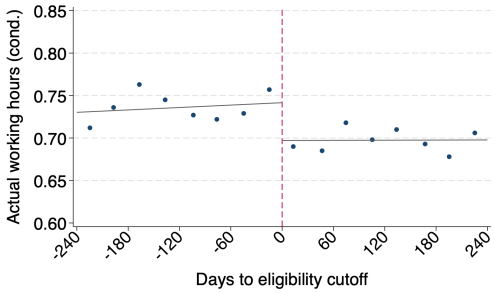
	Contractual working hours (1)	Actual working hours (2)
Panel A: All		
RD estimate	−0.009	−0.043***
	(0.015)	(0.015)
Control mean	0.855	0.855
Observations	5,921	5,921
Panel B: Full-time		
RD estimate	0.023***	−0.023**
	(0.009)	(0.010)
Control mean	0.983	0.983
Observations	2,595	2,595
Panel C: Part-time		
RD estimate	−0.024	−0.048***
	(0.018)	(0.017)
Control mean	0.746	0.746
Observations	3,326	3,326

Notes: The table reports RD coefficients estimated from equation (1). The sample consists of secondary-school teachers employed in 2013 and is restricted to teachers observed in the teacher register in 2016–2018, the period for which leave hours are recorded. *Contractual working hours* are the FTE specified in the employment contract. *Actual working hours* are contractual hours net of subsidized leave, expressed in FTE. Full-time is defined as ≥ 0.95 FTE in 2013; part-time below. The bandwidth is 240 days around the 1 August 1962 cutoff, with a local linear specification and a uniform kernel. Standard errors in parentheses are clustered at the teacher level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

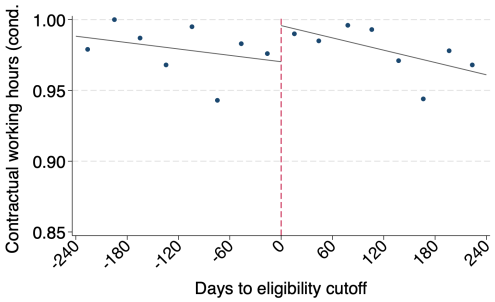
Figure D.1: Effect of eligibility on working hours, by contract type



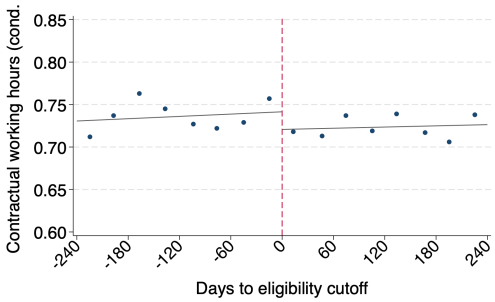
(a) Full-time: actual working hours



(b) Part-time: actual working hours



(c) Full-time: contractual working hours



(d) Part-time: contractual working hours

Notes: Panels (a) and (b) plot actual working hours, defined as contractual FTE net of subsidized leave, and panels (c) and (d) contractual working hours in FTE, against date of birth relative to the 1 August 1962 cutoff, pooled over 2016–2018, the period for which leave hours are recorded. All outcomes are conditional on being observed in the teacher register. Bins are evenly spaced over 30 days with separate linear fits on each side; the bandwidth is 240 days with a uniform kernel. Full-time is defined as ≥ 0.95 FTE in 2013; part-time below.